

A B O O K O F O N E Y E A R

A Year on a Dairy Farm

Twelve Months on a Finger Lakes Dairy

M E I R A D A V I D S O N



For the Jessica family kids. Be curious, stay open, be kind.

FIRST EDITION

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MEIRA DAVIDSON

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Set in Iowan Old Style and Helvetica Neue.

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A Note to the Reader

The people in this book are fictional. Tom and Sarah Hartley, their daughter Anna, Marcelino and his family, the cow named Hazel and the cow named Mabel, the neighbors and the cheesemakers and the truck drivers — none of them are portraits of any one real person. They are composites, built out of years of conversations on grass-fed dairies in the Finger Lakes region of New York, and shaped to fit the rhythm of a year on the page. The farm itself is not any single farm. It is, more honestly, a layered photograph of several.

The single exception is the brief mention of Hank Paulson in Chapter 12. He is a real person, and the small story told there about him and Wendy Paulson — the loft they built together on his parents' farm in Illinois, the wiring she did, the career that happened around the house — is drawn from his published memoir, *On the Brink*, and is included because it gestures at a way of living that very few young readers are told they have access to.

Everything else has been changed, combined, or invented. Any resemblance to a particular farmer, worker, family, or cow is the resemblance one real life always bears to another.

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CHAPTER 1

1

Chapter 01 February Inside The Barn



It is four in the morning, the thermometer outside the kitchen window reads minus six, and Tom Hartley is putting on the second of his two pairs of socks. The first pair is thin wool, the second thick. Then Carhartt overalls over long underwear, then the boots, then a flannel, then the brown duck coat that is missing two buttons because he keeps meaning to fix them and keeps not fixing them. A wool hat. Gloves in the pocket — he won't need them yet, inside.

The kitchen is dark except for the small light over the stove. The coffee is in a thermos; he made it the night before and reheated it just now in the microwave for forty seconds, which is exactly long enough. The radio is on low — the Geneva station, weather and

farm news, a man's voice talking about hay prices in a tone that suggests he has been talking about hay prices for thirty years and intends to keep going. Sarah is upstairs, asleep. Their daughter Anna, home from Cornell for the long weekend, is asleep in her old room. The dog, an aging border collie named Jess who has stopped being useful around cows but is still useful around the kitchen, lifts her head from the rug, considers the situation, and decides to stay where she is.

Tom goes out the back door. The cold is the kind that feels personal. It enters the inside of his nose and turns the small hairs there into a foreign material. He walks the eighty yards from the house to the barn on a path he has shoveled and re-shoveled all week, the snow piled waist-high on either side. The Finger Lakes do not get the heaviest snow in New York — that distinction belongs to the cities along Lake Ontario, Watertown and Oswego, where the lake-effect bands stack up — but they get enough, and they get the kind of cold that comes off the long fingers of water before the lakes freeze and then sits in the valleys for weeks.

The barn comes up out of the dark like a ship. It is a long low building, a free-stall barn, built by Tom's father in 1978 and added onto twice since. The east-facing windows of the milkhouse are already lit. Marcelino has been here since a quarter to four. He is from a town in the highlands of Guatemala that Tom has looked up on a map exactly once, in 2019, when Marcelino first came, and which Tom could not now find again without looking again. Marcelino speaks Spanish to Tom and Mam to his wife on the phone, and the phone is in his shirt pocket, on speaker, more often than not, because his daughter, who is seven, likes to hear what is happening even when there is nothing to say.

Tom slides open the door to the milkhouse and the warm hits him in the face. It is not warm warm. It is forty-five degrees, maybe forty-eight, and after minus six this is the tropics. The milkhouse smells like detergent and stainless steel and faintly of milk, which is the cleanest smell in any working dairy and the one that, if you have ever lived around it, you can identify in a stranger's coat from across a room. Marcelino is at the wash sink, hosing down a length of pipeline. He nods. *Buenos días. Buenos.* The vacuum pump is already running, a steady low hum that you stop hearing after about a minute and then notice again, like the refrigerator at home, only when it cuts out.

Through the next door is the parlor — a double-six herringbone, which means twelve cows can be milked at a time, six on each side, standing at an angle to the pit where Tom and Marcelino will work. The pit is a trench three feet lower than the cow platform, so the cow's udder is at chest height for a person of average build, which Tom is and Marcelino is not; Marcelino works on a small wooden box he made himself, two winters ago, and which he has carried into the pit every morning since. The pit smells like rubber and warm metal and very faintly of the iodine teat dip in its bottles along the wall.

Beyond the parlor is the barn proper. This is where you would go if you wanted to understand what a dairy farm in February actually is.

It is two hundred feet long and seventy feet wide and the roof is high, and the cows are in it. There are ninety-four of them this morning. The number floats between eighty and ninety-five over the year, depending on who is fresh and who is dry and who has gone on the truck. They are lying in stalls bedded with sawdust, or

standing at the feed bunk pulling at the long ropy strands of baleage — last summer's grass, cut and wrapped wet in white plastic and fermented into something between hay and silage — or walking, slowly, in the alley between the rows of stalls. The barn is open at the eaves; in summer the breeze moves through it; in winter the warmth of the cows themselves, ninety-four bodies of about fourteen hundred pounds each, keeps the inside of the building twenty to thirty degrees warmer than the outside. You walk in and you can feel them before you see them. You can hear them: the low rumbling of cud chewing, which is constant, and the occasional deep exhale, and the sound of a hoof on the concrete alley, and the wet pulling sound of a tongue wrapping around hay.

The smell is not what people expect. People who have never been in a well-run dairy barn expect it to smell like manure, and there is manure — there is always manure, it is the by-product of the entire operation and a dairy farm is a manure-management business that happens to also produce milk — but a clean barn does not smell bad. It smells like grass that has been turned into another thing. It smells, faintly, sweet. There is the musk of the cows themselves, which is its own warm animal smell, and the diesel of the skid steer parked near the door, and the ammonia note that rises when fresh manure hits old straw, but underneath all of it is the grass, fermented and broken down and passing slowly through the ninety-four enormous four-chambered stomachs that surround you in the dark.

The lights are on, but only the work lights — long fluorescent tubes down the center alley, leaving the stalls in a softer half-dark. The cows do not need the brightness, and Tom does not want to startle the ones who are still lying down. They will get up in their own time, when the parlor calls them.

He walks the length of the barn. He does this every morning and he is doing it now: a slow walk down the feed alley, looking at each cow he passes, not in any deliberate way but in the way a person looks at the rooms of his own house when he comes home at night. He is checking, without quite knowing he is checking, that everything is the same as it was. A cow lying down with her head turned around against her flank: normal, she is sleeping. A cow standing alone at the back of a stall, not eating, ears low: not normal, look again. A cow with her tail lifted: about to do one of two things, both of which Tom has seen ten thousand times.

He passes the milking string — the cows currently in production, the ones who will go into the parlor in twenty minutes — and continues to the back of the barn, to a smaller pen separated by a metal gate.

This is the dry pen. There are eleven cows in it. They are the cows who are pregnant and close to calving, who have been taken off milking for the last two months so their bodies can rest and put their energy into the calf instead of into the bucket. They are fed differently from the milking cows — less rich, more roughage — and they are watched more carefully, because this is when things go wrong, and when things go right.

In the back corner, lying down, is a Jersey-Holstein cross with a black-and-white face and a soft brown muzzle, ear tag number 47. Her name is Hazel. She is six years old. She has had five calves

already and she is three weeks from her sixth. Her belly, even lying down, is enormous; you can see, if you stand still long enough, the calf inside her shifting. Tom leans on the gate and watches her for a minute. She watches him back, calmly, the way she has watched him every morning of her life.

This is where the year begins. Not in March, when the calf will come. Here, in the dark of a February barn, with a pregnant cow and a man leaning on a gate and the vacuum pump humming through the wall and a worker in the next room rinsing a pipeline and a farm and a family and a small economy of milk and grass and labor and weather all waiting, quietly, for the thing that is about to happen.

It is four-twelve in the morning. The first cows are getting up.

The first group of twelve comes in on their own. The parlor has a return lane that loops around behind the milking pit, and the cows know the route the way you know the route from your bedroom to the bathroom in the dark. They walk in, find their stanchions, lower their heads to the small pile of grain in front of each one — the only bribe the system requires — and stand still. Tom and Marcelino move down the line. A spray of pre-dip on each teat, a wipe with a clean towel, attach the milking unit, move to the next cow. The unit pulses and pulls; the milk runs through clear lines into the bulk tank in the next room, where it is chilled within seconds to thirty-eight degrees. Twelve cows take about eight minutes, start to finish. There are seventy-six cows in the milking string this morning. Six and a half rounds. They will be done by seven-thirty.

This is the compressed version. Another chapter will tell it slowly. For now: it is a rhythm, and the rhythm is the same as it was yesterday and the same as it will be tomorrow, and there is a kind of comfort in this even when your back hurts, which Tom's does, in the same place it always hurts, just below the right shoulder blade, where he tore something thirty years ago lifting a hay bale he should not have lifted alone.

By eight he is back in the kitchen. The light is up now, the kind of February light that doesn't commit to being daylight — gray, low, indirect, like the sky is making a half-effort. Sarah is at the stove. Anna is at the table in pajama pants and one of Tom's old flannel shirts, drinking coffee and looking at her laptop. Jess is under the table.

"How's Hazel?" Sarah says.

"Fine. Big."

"Soon?"

"Soon."

Anna looks up. "Is it Hazel? The one I helped pull out of the mud that time?"

"That was Hazel."

"How old is she now?"

"Six."

Anna considers this. "She's old."

"She's not old."

"For a dairy cow she's old."

Tom sits down. Sarah puts a plate in front of him: two eggs, toast, bacon she got at the farmers market in Trumansburg. He picks up his fork. "She's not old," he says again, to no one.

Sarah laughs, the small private laugh of a wife who has been listening to a particular argument inside her husband's head for thirty-two years.

After breakfast the day breaks open into the things that have to be done, and the things that have not yet decided whether they have to be done. The first category includes feeding the dry cows, refilling the calf hutches with fresh straw, pushing up the feed at the bunk so the cows can reach what they had pawed away from themselves overnight, and starting the skid steer to scrape the alleys. The second category is what makes a farm a farm: the unscheduled thing that happens at 9:47 in the morning when you are halfway through scraping an alley and Marcelino comes out of the heifer barn and says the waterer is frozen.

There are two waterers in the heifer barn. They are heated. They are heated with a small electric element that has, for reasons the manual does not fully explain, decided last night not to heat. Now there is a domed cap of ice over the surface of one of them, and the heifers — six of them, between four and twelve months old — are standing around it looking at it and at Tom with the particular patience of animals who have understood that a problem exists and that it is not their problem to solve.

Tom goes back to the shop for a hammer and a kettle and an extension cord. The hammer breaks the ice. The kettle, plugged into the extension cord, pours hot water on the element. The

element comes back on, eventually. The whole repair takes forty minutes and accomplishes nothing that wasn't already supposed to be happening on its own. This is a lot of farming.

At noon Marcelino goes back to the trailer. The trailer is a single-wide set up on the slope behind the barn, on cinder blocks, with the propane tank visible from the kitchen window of the main house. Tom's father put it in for a hired man in 1991 and it has been a hired man's home ever since, with different hired men in it. Marcelino has been in it for almost five years. The inside is clean — Marcelino is meticulous about this — and the kitchen has a small electric kettle and a pan and a bag of dried corn for tortillas and a stack of plates with a pattern of yellow flowers around the rim that he chose at the Walmart in Geneva because his daughter had picked out the same pattern, in a smaller plate, for her grandmother in Huehuetenango.

He eats lunch alone most days. Today he eats lunch alone. He puts the phone on the counter on speaker, and his wife answers, and for a while there is the small clatter of two kitchens — one in upstate New York and one in the highlands of Guatemala — and the voice of his daughter, who has just gotten home from school, asking him if it has snowed today.

Sí, hija. Mucho. Mucho mucho.

¿Te divertiste?

He laughs.

In the afternoon Tom does the things that involve paper. He does not like the things that involve paper. He has done them his entire adult life and he does not like them yet. Sarah is at the kitchen table with the laptop and a stack of milk receipts, three

months of them, and she is entering numbers into a spreadsheet and frowning at the screen the way she used to frown at her students when she was a fourth-grade teacher in Penn Yan and did not believe their excuses about their homework.

"How was last month," Tom says, getting coffee.

"It was last month."

"That bad?"

"Not bad. Last month."

Tom looks at the spreadsheet over her shoulder. There is a column for income and a column for expenses and a column for difference. The difference column has both kinds of numbers in it. He does not stay long enough to count which kind there are more of. He has the kind of relationship to this spreadsheet that you have to a check engine light: it is on, you know it is on, you will deal with it, you do not need to look at it right now.

"Is Anna still here for dinner?"

"She's leaving Sunday."

"Good."

He goes back outside.

The afternoon milking starts at three-thirty and ends at six. It is the same as the morning except that the light is failing instead of arriving, and the cows are slightly slower, and Tom is more tired, and Marcelino does not say much, and there are no eggs at the end of it. By six-fifteen Tom is back in the barn pushing up feed for the night. By six-thirty the cows are settling.

He does the last walk-through at twenty to seven. The barn is quiet. The cows in the milking string are lying down or eating. The dry cows are at the back. He walks the alley slowly, the way he walked it that morning, looking at the ones he passes.

Hazel is standing now, at the bunk, pulling at a rope of baleage with the steady patient work of a cow who has thirteen hours of digesting ahead of her and intends to begin. Her belly hangs low. Her ears are forward. She does not turn to look at him. She is busy.

Tom stands at the gate and watches her for a minute.

Outside, the wind is moving across the snow. Inside, it is still warm. The vacuum pump is off. The fluorescent lights are off; only the small night lights at the end of each row are on, casting the barn in a low blue. The cows breathe in the half-dark. Somewhere a hoof shifts on concrete. Somewhere a cow lets out a long, slow exhale, the kind that means she is settling for the night.

In three weeks, give or take, Hazel will have a calf.

Tom closes the gate and turns out the last of the lights and walks back across the snow to the house, where Sarah is making dinner and Anna is on the phone and Jess is under the table and the radio in the kitchen is still on, very low, talking now about something other than hay.

CHAPTER 2

2

Chapter 02 The Cow Herself



You cannot eat grass.

Try it. Pull a stem out of the lawn this summer and chew on it for a while and notice what happens. You will get a faintly green taste, a slightly bitter edge, a stringy feeling in your teeth, and at the end of it, exactly zero calories that your body knows what to do with. The energy is in there. You can see it — the grass is alive, growing, photosynthesizing, doing chemistry that human cells cannot do. But the calories are locked behind walls of cellulose, which is a molecule your digestive system cannot break apart. You could eat grass for a month and starve.

A cow eats grass. So does a sheep, a goat, a deer, a moose, a giraffe, a bison. They are called ruminants — about two hundred species in all — and they have figured out a thing the rest of the

mammal world has not. Their solution is one of the strangest tricks in biology, and it is the reason humans built civilizations on top of cattle. A cow can take what is growing on land we cannot farm — hillsides, scrub, prairie, marshland, the back forty acres of a steep New York farm where you'd never get a tractor — and turn it into protein and fat that we can eat.

She does this with bacteria.

A cow has four stomach chambers, but only the last of them — the abomasum — is anything like the stomach inside you. The other three are something else. The largest of them, called the rumen, is a fifty-gallon vat of warm fluid that takes up most of the left side of her body. If you put your hand on a cow's flank just behind her ribs you can feel it: a slow continuous churn, like a very large slow-motion washing machine. The rumen is packed with microbes — bacteria, protozoa, fungi — billions of them in every drop of fluid. They do something a cow cannot do on her own: they break cellulose down. They eat the grass that the cow swallowed. They reproduce inside her. And then, eventually, the cow digests them.

The cow is not digesting grass. The cow is digesting the bacteria that digested the grass. She is a farmer of microbes inside her own body, and the entire arrangement is a partnership a hundred million years older than we are.

The four chambers do four jobs. The rumen is the fermentation tank. The reticulum, smaller and sitting next to it, catches the bigger pieces and sends them back up the cow's throat to be chewed again. This is the cud. When you watch a cow lying down in the afternoon with her eyes half-closed, jaw moving sideways in that slow circular motion that looks almost meditative, what she is

doing is regurgitating a wad of partly-fermented grass and re-chewing it to break it down further. If the barn is quiet, you can sometimes hear it: a soft wet click, every minute or so, as the next wad comes up. A cow chews her cud about eight hours a day. A healthy cow. A cow who isn't chewing is a cow who is sick. The third chamber, the omasum, is a folded sieve that squeezes water out of the mass before it passes to the abomasum, the true stomach, where it finally meets enzymes and acids the way food meets your stomach when you eat dinner.

All of this — the four chambers, the microbes, the eight hours of chewing — is in the service of one outcome. Grass goes in. Milk and meat come out.

We did not invent this animal. We modified one.

The cow's wild ancestor was a creature called the aurochs. The aurochs lived across Europe and Asia and North Africa for hundreds of thousands of years. She was bigger than a modern cow — six feet at the shoulder, sometimes more — and she was dangerous. She had long forward-curving horns and a temperament that did not invite handling. The aurochs was the wild ox of the cave paintings at Lascaux. She was hunted by lions and by saber-toothed cats and, later, by us.

About ten and a half thousand years ago, in two places — somewhere in what is now Turkey or Iran, and separately in the Indus Valley in what is now Pakistan — small groups of humans began doing something different. Instead of hunting aurochs, they captured them. They kept them alive. They kept the calmer ones and ate the wilder ones, and let the calm ones breed. Over generations, the descendants of those animals got smaller, gentler, more pliant. They got coats in colors no aurochs would have worn.

They got useful: they pulled plows, they carried loads, they gave milk. The wild aurochs stayed wild, alongside, for thousands of years more, getting rarer and rarer as forests were cleared and herds were squeezed. The last aurochs on earth was a female who died in a forest in Poland in 1627, alone.

What was left of her, by then, was already living on every continent humans had reached. We had remade her into the cow.

There is no such thing as “the cow.” There are breeds. Hundreds of them. Most readers of this book will know two or three by sight without knowing they know.

The big black-and-white cow — the one on the milk carton, the one in the picture book, the one a six-year-old draws when you say *draw a cow* — is a Holstein. She is the volume producer, the workhorse of American dairy. About 90 percent of the milk in your refrigerator came from a Holstein. She is very large by cow standards, fifteen hundred pounds or more, and she gives an astonishing amount of milk: about eight gallons a day on a commercial farm, sometimes twelve. She came originally from the lowlands of the Netherlands and northern Germany.

The smaller brown cow with the deer-like face, the long eyelashes, the dark muzzle — that’s a Jersey. She comes, originally and exactly as her name suggests, from the island of Jersey, off the coast of France. She is half the size of a Holstein, gives less milk, but the milk she gives is much richer in butterfat and protein, which is why if you buy fancy butter or fancy cheese or fancy ice cream and read the label you will often find the word *Jersey* somewhere on it.

Then there are the others, fewer in number, more particular. The Brown Swiss, gray-brown and gentle, originally from the Alps, known for living a long time. The Guernsey, golden-yellow, also from the Channel Islands, whose milk has a slightly creamy color from extra beta-carotene. The Ayrshire, red-and-white, from Scotland, hardy on hills. The Milking Shorthorn, an English breed that used to be everywhere in America before Holsteins took over. The Dutch Belted, black with a wide white stripe around her middle, of which only a few thousand remain in the world.

A grass-based organic farm in the Finger Lakes is unlikely to have a herd of pure Holsteins. The Holstein was bred for industrial milk production indoors on rich grain rations. On grass, she struggles. She is too big, eats too much, breaks down too quickly. So farms like Tom's tend toward smaller cows — Jerseys, or crosses. Hazel is a Jersey-Holstein cross: half the genetic legacy of the lowland Dutch volume cow, half the legacy of the small rich-milk island cow. She gives less than a Holstein and more than a pure Jersey. She is hardier than either. She holds up on grass.

The other thing to know about modern cows is how recently they became modern.

In 1950, the average American Holstein produced about five thousand pounds of milk in a year. Today the average American Holstein produces about twenty-five thousand pounds. She gives, in other words, five times as much milk as her great-great-grandmother. This did not happen because cows changed their minds about how much milk to make. It happened because for seventy-five years we selected, every generation, the daughters of the highest producers and bred them to the sons of the highest

producers, and over time we produced an animal whose body has been pushed in one direction further than any farm animal in history.

The cost of this is real and worth knowing. The high-production modern Holstein has more difficulty getting pregnant than her ancestors. Her feet are more likely to give out under her weight. Her udder is more prone to infection. Her metabolism, in the months after calving, runs at a rate that pushes her body into a kind of controlled depletion: she gives so much milk that her body cannot eat fast enough to keep up, and she draws down her own reserves to do it. She is an animal we have engineered to operate at the edge of what her biology can sustain.

This is one reason farms like Tom's choose smaller, mixed-breed cows. A Jersey-Holstein cross gives less milk than a champion Holstein. She also lasts longer.

Which brings us to the question of how long a cow lives.

A cow's natural lifespan — the number of years she would live if she were left to be a cow, without being bred for production, without being sent anywhere at the end of it — is about eighteen to twenty-two years. There are records of cows reaching their late twenties. They are the exceptions. Eighteen to twenty-two is the working number.

The average dairy cow on a conventional American farm is culled — taken out of the milking herd and sent to slaughter — at about five years old. She has had three lactations. She is, in human terms, the equivalent of someone in their early twenties.

On a grass-based organic farm, the average is older. Maybe seven, eight, nine. Hazel is six. She has had five calves. If she stays healthy and keeps breeding back and keeps producing well, she might be on the farm for another three or four years. If something goes wrong — if she doesn't breed back, if her feet give out, if her production drops — she will not.

This is part of what a dairy cow is. It is not the whole of what she is. It is also not nothing.

Now: what is her day like?

Most of what we know about how a cow experiences the world we owe to a small number of researchers, but more than any other to a woman named Temple Grandin. Grandin is autistic, which she has said gave her a kind of access to cattle thinking that most people don't have. She thinks in pictures, specifically and concretely, the way she has argued cows do. She has spent fifty years in feedlots and stockyards and slaughterhouses watching what makes cattle calm and what makes them panic, and she has redesigned much of the American cattle handling system around what she found.

What she found is that cows perceive the world in ways that are not like ours, and that almost everything that frightens or stresses a cow can be predicted if you know what to look for.

A cow's eyes are set on the sides of her head, not the front. This gives her more than three hundred degrees of vision around her body — almost everything except a narrow blind spot directly behind her tail. This is what an animal who has been hunted for millions of years needs in order to survive. The price of the wide field of view is poor depth perception. A cow cannot easily tell

whether a dark patch on the floor is a shadow or a hole. She will hesitate. If you push her, she will refuse, and you will think she is being stubborn. She is not being stubborn. She is being a cow.

A cow's hearing is more sensitive than ours, especially at high frequencies. The clang of a metal gate, the squeal of a hydraulic pump, the high whine of a truck engine — all of this is louder to her than to you. A cow startled by a sudden sound will remember that sound, and that place, for years. A cow who got jabbed in a particular alley three years ago will balk at that alley today.

Her sense of smell is far better than ours. She knows the smell of her own herd, of her own calf, of the people who handle her. She can smell stress on other cows.

She sees in color, but not the same range we see; there is some debate. She is sensitive to contrast — a yellow rain coat hanging on a fence will spook a herd that has walked past that fence a thousand times. She thinks in specifics. She does not generalize. The yellow coat is a problem. The green coat that hung there yesterday was not.

She has friends. This is not sentimental language. Cows form preferred social bonds with specific other cows. They graze together, they lie down next to each other, they get distressed when separated. There is a hierarchy in any herd — a top cow, a bottom cow, a working order in between — and the hierarchy is stable over years and remembered by every member. When a new cow joins the herd, she will be challenged, sometimes hard, until her place is sorted out. Once sorted, the herd settles.

She has a memory longer than people once thought. She remembers individual humans. She remembers which ones moved her gently and which ones did not. She remembers the shape of her own barn and the route to her own stanchion in the milking parlor. If you change the layout of the milking parlor on a Monday morning, you will pay for it for a week.

What this adds up to is a creature whose inner life is real and specific and largely closed to us. She is not like a dog; she does not love you. She is not like a wild animal either; she has been remade by us. She is something else. A cow.

If you stand at the gate of the dry pen at six in the evening and watch Hazel pulling at a rope of baleage, all of this is in front of you. A fifty-gallon fermentation tank turning grass into milk. Ten thousand years of selection from an animal whose last wild ancestor died alone in Poland. A breed shaped by an island off the French coast, crossed with a breed shaped by a Dutch coast. A body engineered, over the last three generations, to do something no cow before 1950 ever did. A nervous system that sees almost all the way around itself, that hears the milking pump from the next room, that has memorized the route from her stall to the parlor and would walk it in the dark. A six-year-old animal who has had five calves and is about to have a sixth, and who might live another four years on this farm. Or longer. Or not.

She does not know any of this. She is busy. She is eating.

CHAPTER 3

3

Chapter 03 March A Calf Is Born



It is two-fourteen in the morning, and Tom is at the gate of the calving pen with a mug of coffee he is mostly not drinking, and Hazel is going to do this on her own.

He came out at one-thirty. She had given him the look. The look is something a farmer who has watched a hundred cows calve learns to read without being able to explain — a way of holding the head, separated slightly from the herd, ears not quite forward, eyes turned more inward than outward. At nine-thirty the night before, on his last walk-through, he had seen it on her, only just, and known. He went to bed in his clothes. He set the alarm for one. He woke up at twelve-fifty without it.

Now he is here, in the half-dark of the calving pen, which is a separate pen at the back of the barn deep with fresh clean straw, the kind that smells faintly sweet and crackles when you walk on it. There is one cow in the pen. Hazel is lying down on her side. Her legs are stretched out. Her tail is up. There is a clear glistening sac the size of a basketball emerging from her, and inside it, you can see two small front hooves.

This is what is supposed to happen.

A cow is pregnant for two hundred and eighty days. Which is to say a cow's pregnancy is three days shorter than a human's, almost exactly, which feels like a fact someone made up but is not. Over those nine and a third months a calf grows from a single fertilized cell to an animal weighing seventy or eighty or ninety pounds, with a working circulatory system, a working nervous system, four chambers of stomach folded up and waiting to inflate, a tongue, a coat, eyelashes, hooves. It grows inside the warm dim country of the cow's body, fed by her blood, breathing through her placenta, suspended in fluid that is the same temperature as her insides.

When it is ready to come out, several things happen at once. The cow's cervix begins to dilate — a slow soft opening that can take six hours or more, and during which she will stop eating, stand and lie down repeatedly, walk to the back of the herd, look at her flank as though checking on someone behind her. This is the hours-long thing Tom watched at nine-thirty and called *the look*. After dilation, what is called active labor begins, and the calf, who has been positioned head-first inside the cow for the last few weeks of pregnancy, starts moving toward the world.

The first thing to come out is not the calf. It is a sac of fluid — the amniotic sac, the inner of the two membranes that have been holding the calf — and it appears at the cow's vulva like a translucent gray-pink water balloon. This is what Tom is looking at now. Inside it, a small dark shape is visible: two front hooves and the tip of a nose, side by side, the way the calf is meant to come.

A normal calf comes nose-first, front feet first, like a swimmer diving off a board. The feet appear, then the nose tucked between them, then the head, and once the head is out the rest of the calf comes quickly, in one long slippery slide, because the shoulders are the widest part of him. If the calf is positioned wrong — if his head is bent back, if he is coming backwards, if a leg is folded under — the farmer has to intervene. Tom has done this. He has reached up inside a cow with a chain and a glove and a steady arm and turned a calf and pulled him through. He is good at it. He does not enjoy it. Tonight he will not need to do it.

Hazel pushes, and the front legs come further. She pushes, and the nose appears.

It is two-thirty-one when the calf is fully out, and he is enormous, and he is alive, and he is — Tom can see this in the half-light, even before Hazel has stood up — a heifer. A female calf.

For the next thing in the book to make sense, you have to know what would have happened if she had not been.

About half of all calves born on a dairy farm are male. They cannot become dairy cows. A male calf grows up to be a bull, and a bull does not give milk. So on every dairy farm in the world,

every day, in equal numbers to the female calves who will replace their mothers in the milking string, male calves are born who have no place in the dairy system at all.

What happens to them depends on the farm. On the worst farms, in some parts of the world, male calves are killed shortly after birth. This is not typical in the United States, but it happens. On most American dairy farms, male calves are kept for a few weeks, fed milk replacer, and sold to a veal operation or a beef operation. On a farm like Tom's, male calves stay in a hutch for the first three or four months alongside the female calves, are fed the same way, and then are sold to a local beef farmer who raises them on grass to about eighteen months and sells the meat. Tom does not name the male calves. The female calves get names because they are going to be on the farm for years; the male calves get tags, because they are not.

This is part of the system. It is not hidden. It is just how it works. There is no version of dairy — no organic version, no grass-fed version, no European version, no small-farm version, no version anyone has yet figured out — in which half the calves born do not need to be male. The cow has to be pregnant to give milk; pregnancy makes calves; half the calves are male; the male calves do not give milk. That sentence is the entire arithmetic.

The calf in the straw in front of Tom is a heifer. She will stay on the farm. In two years she will have her first calf and join the milking string. She is, in the language of the farm, a *replacement* — a cow being raised to replace, eventually, a cow who is no longer here.

But all of this is for later. Right now she is wet and steaming in the cold air and she has just been born.

Hazel rolls onto her chest. She rests for a moment. Then, in one slow heaving motion, she stands up, and the umbilical cord, still attached, breaks where it is supposed to break, and a little blood lands in the straw, and Hazel turns around and begins to do what cows do.

She licks her calf.

She does not stop licking her calf for forty minutes. She licks her starting at the head — the wet sticky head — and works downward, methodically, with a tongue the size and shape of a thick wet washcloth. She licks the membrane off the calf's face. She licks the calf's eyes open. She licks the calf's neck, her shoulders, her flanks, her belly, her legs. She turns the calf over with her nose so she can lick the other side. The calf is breathing. The calf opens her eyes. The calf does not yet know what eyes are.

Tom watches from outside the gate. He does not go in. There are things he will need to do — he will need to dip the calf's navel in iodine to prevent infection, he will need to make sure the calf gets colostrum, the rich first milk that contains the antibodies the calf needs to survive her first weeks of life — but he does not need to do any of them yet. Right now Hazel is doing what Hazel was made to do. The licking is not just affection, though it is also affection. The licking dries the calf, stimulates her circulation, warms her, helps her breathe. The licking is what the calf needs in order to live, and Hazel knows it without knowing she knows it.

It is two-fifty-six. The calf, still wet, still mostly Hazel's tongue, lifts her head.

A calf, if she is healthy, stands within the first hour of life. She does this on her own. She does it badly at first. It takes her several attempts, with much sliding and folding of legs and unexpected sitting down. Hazel keeps licking her. The calf, in the middle of being licked, decides to try standing again. Her front legs come out first. She gets her head up. She gets one back leg under her. The other back leg gets confused. She falls over.

This is normal. Tom does not help.

By three-fifteen the calf is standing, swaying, her head down, on legs that look impossibly long for the rest of her. By three-twenty she has taken her first step. By three-twenty-five she is looking for the udder, by smell — she does not know what an udder is, but she knows a thing about her mother's body draws her — and she finds a teat, and she latches.

The first milk a cow makes is not regular milk. It is colostrum: thick, yellow, sticky, incomparably rich in antibodies and growth factors and nutrients. The calf is born with no immune system to speak of. She gets her immune system from the colostrum, in the first few hours of life, through a window that closes by about twenty-four hours. After that, her gut can no longer absorb the antibodies whole. Tom will, in a little while, milk Hazel separately and bottle-feed the calf to make sure she gets the full amount; that is the modern dairy practice, and it has measurably reduced calf mortality. But this first drink, right now, in the dark, from her mother — is its own thing. Tom watches for a few seconds and looks away.

He goes inside the pen, kneels in the straw, dips the calf's umbilical stump in a small cup of iodine. The calf does not seem to notice. Hazel does not seem to notice. He walks back out and closes the gate.

It is three-thirty-eight. He is going to bed.

At seven the next morning Anna comes down to the barn in a pair of his old rubber boots and her grandmother's coat. Tom is back in the parlor with Marcelino. The cows have been milked. The barn is in its full daytime version of itself, which is a different country from the country of three a.m.

"Did she come?"

"She came."

"Is it —"

"Heifer."

Anna's whole face lights up the way it has been lighting up since she was four. She walks down to the calving pen. Hazel is standing at the bunk, eating, the long ropy strands of baleage sticking out of her mouth, the placenta passed and removed, her tail crusted with the small dried evidence of the night. The calf is in the straw, dry now, brown-and-white, blinking, awake, exhausted. Anna kneels at the gate and looks at her for a long time.

"What's her name?"

This is a thing Tom does for the heifer calves born in the night when Anna is home: he lets her name them. He has done it since she was eight.

"Up to you."

Anna thinks about it. The calf looks back at her with the slow unfocused gaze of a creature who arrived in the world four and a half hours ago and is still working out what most of it is.

“Mabel.”

“Mabel,” Tom says.

He writes it down in the herd book that afternoon. *Mabel. Born March 14, 2:31 a.m. Dam: Hazel #47. Sire: 7HO15428.* The sire number is the bull whose semen Tom paid for last May. Mabel’s father is a bull who lives in Wisconsin and whom none of them have ever seen and who has, by genetic arithmetic, several hundred daughters scattered across small farms in the upper Midwest and the Northeast that he also has never seen.

For now: Mabel.

By the next evening Mabel is in a hutch.

A hutch is a small individual outdoor pen — a fiberglass dome the size of a doghouse, attached to a wire pen about six feet square, bedded deep in straw — and on Tom’s farm there are eight of them in a row along the south wall of the heifer barn, where the sun warms them. Mabel will live in hers for the next eight weeks. She will be fed twice a day from a bucket: pasteurized milk that comes from her mother and the rest of the milking string, warmed to the right temperature, delivered by Sarah or by Marcelino or by Tom. She will not nurse from Hazel again.

This is the part of dairy that the smartest readers of this book will already have known was coming. It is the part that is hardest to write warmly because there is nothing warm about it. The calf and the cow are separated. They can hear each other for a day or two — Hazel will bellow for Mabel from the milking string; Mabel

will bellow back from the hutch. After a few days they stop. Within a week Hazel is back in her routine. Within a month Mabel has bonded to whichever person feeds her morning and night, and that person is a substitute for what she briefly had.

Why is this done? There are reasons. Some are good. Some are pragmatic. The good reasons are about disease — the calf's gut and immune system are vulnerable in the first weeks, and individual housing reduces transmission — and about ensuring she gets the right amount of colostrum, which is much harder to confirm if she is nursing free-form. The pragmatic reason is that the cow's milk is what the farm sells. A calf nursing from her mother for twelve weeks would drink several thousand dollars' worth of milk over that time, and the farm would not exist.

There is a small but growing movement of dairy farms — some in Europe, a few in the U.S. — experimenting with cow-calf dairying, in which the calf stays with the mother for months. The early data suggest both cow and calf are happier. The economics are very hard. Tom knows about it. He has thought about it. He has not done it.

What he does instead is what most small grass-based organic farmers do. He keeps Mabel in a clean dry hutch where she is well-fed and well-cared for, and he tells himself, accurately, that she is treated better than calves on most American dairy farms, and that this is true does not change the other thing that is also true, which is that she is not with her mother.

It is dusk on the second day. Tom does the evening walk-through. The barn is settling. The milking string is back in the stalls. The dry cows are in their pen at the back. Hazel is among the milkers now, eating, her udder full and beginning the work it will do for the next three hundred days.

Out behind the heifer barn, in the eight hutches in a row, eight calves are settling for the night. Mabel is the newest of them. She is curled in a half-circle in the deep straw of her hutch, her head tucked against her flank, her eyes closed. The straw is warm from her body. She is two days old.

Tom stops at her hutch for a moment and watches her sleep.

"Mabel," he says, quietly, to no one.

Then he closes the gate of the heifer yard, and walks back across the pasture lane in the gathering dark, toward the house, where the lights are on and Sarah is making dinner and Anna is at the kitchen table doing homework she should have done two days ago. The barn is quiet behind him. The cows breathe. Somewhere in the middle of the milking string, Hazel chews her cud.

It is a beginning.

CHAPTER 4

4

Chapter 04 April Out To Pasture



It is the third week of April, and the grass is six inches tall, and the ground is firm, and the forecast is for sun, and Tom Hartley has decided that today is the day.

He has been deciding for a week. The grass had been ready, more or less, since the previous Wednesday. The ground had been the question. After a winter of forty inches of snow and a spring of cold mud, you cannot put ninety-four cows on a pasture that has not finished firming up, or they will punch their hooves a foot deep into it and turn the field into something you cannot fix in one summer. So Tom has been walking the lower pasture every morning at six-thirty with a stick, pressing it into the ground and pulling it back out and looking at how much soil came up with it.

By Tuesday the stick came back clean. By Wednesday he was sure. He waited two more days, because there is no benefit to being early and there is real cost to being early, and now it is Friday, and the grass is six inches tall and the ground is firm and the forecast is for sun.

The cows have been inside since November.

The barn doors are wide open. Marcelino is at the back of the milking string moving cows forward; Tom is at the gate. The cows know something is up — they always do, on this morning, every year, even though some of them have done this only twice in their lives and some of them have never done it at all — and they are walking with their heads up and their ears forward and a kind of low murmur is moving through the herd that you do not hear in the barn in February.

The first cow out of the gate is an old Brown Swiss named Bertha who has done this seven springs in a row, and what she does, having stepped onto the path, is walk twenty feet and put her head down and start eating, immediately, without ceremony, like a person who has been waiting for the line at the buffet to clear. The next three cows do the same. The fifth cow, younger, hesitates at the edge of the pasture, sniffs the air, takes one step, sniffs again, takes another step, and then, for reasons that are not entirely her own to explain, leaps straight up into the air with all four feet off the ground at once.

Then all of them go.

Ninety-four cows, fourteen hundred pounds each, running. They run in loose loops around the pasture, kicking out their back legs, tossing their heads, banging into each other, doing things

with their bodies that you would not have guessed cows could do. Hazel comes out of the gate at a trot and accelerates. Her udder, full from the morning milking, swings under her. She runs the way Mabel ran across the floor of the calving pen on the morning she was born — with the same unsteady joy, only at fourteen hundred pounds.

It takes about three minutes. Then it is over. Then ninety-four cows have their heads down and are eating, scattered across two acres of grass that until five minutes ago had not been touched by a cow in five months.

Tom and Marcelino lean on the gate.

“Cada año,” Marcelino says, smiling.

“Cada año.”

What you are looking at, when you are looking at a pasture, is not what you think you are looking at. You are looking at the top of something.

The grass — the green, the part the cow eats — is maybe a third of the total living mass of a healthy pasture. The other two-thirds is underneath. Roots, going down four or six or eight feet. Microbes — bacteria, protozoa, fungi — a billion or more of them in every gram of soil. Mycorrhizal fungi making thread-thin networks through the dirt and connecting the root of one plant to the root of another in a kind of slow underground internet, through which the plants trade water and nutrients and sometimes warnings. Earthworms turning the soil over. Beetles and springtails and nematodes. Roots dying and being decomposed and adding their carbon back to the soil. The pasture is an ecosystem, and the grass on top is what it produces.

A pasture builds soil. A cornfield depletes it. This is the difference, ecologically, between a pasture-based dairy and the kind of farm that grows corn and soy to feed cows kept indoors in confinement. The cornfield is plowed, fertilized, planted, sprayed, harvested. The soil is disturbed every year. Organic matter is mined out. After several decades of corn-soy rotation the soil is thinner and biologically simpler than it was. After several decades of well-managed pasture the soil is deeper and biologically richer than it was. Pasture is one of the few forms of agriculture that adds back to the land it uses.

When you stand in a pasture in April and the cows are eating, what is happening underneath you is that the grass, sensing that some of it has just been bitten off, is signaling its roots to push more sugars down into the soil. Some of those sugars feed the mycorrhizal fungi the grass is partnered with. Some of them feed the bacteria around the roots. Some of them are exuded directly into the soil to feed whatever is hungry. The grass is gardening. It is doing this because over millions of years grasses evolved with grazers, and grass that is grazed and then allowed to recover comes back stronger than grass that is left alone. The cow biting off the top of the grass is part of how the grass intends to live.

This is the deepest fact about a grass-based dairy, and it is the one that no marketing copy can communicate. The cow and the grass are not in a transaction. They are in a partnership a hundred million years old.

What grows in a Finger Lakes pasture in April is not one plant. It is many.

If you walked Tom's lower pasture on a sunny day in May with a notebook, you could probably identify between fifteen and twenty-five different plants in the space of a single acre. Some of them are grasses — orchardgrass, timothy, tall fescue, Kentucky bluegrass, a little perennial ryegrass. Some of them are legumes, which is the family of plants that includes peas and beans and which has the special biological trick of pulling nitrogen out of the air through bacteria living in nodules on their roots — white clover, red clover, birdsfoot trefoil. Some of them are what most people would call weeds, and which are nutritious forage that cows like — dandelion, plantain, chicory. A good pasture has all of this growing together. The cows, given the choice, will eat some of each, and the milk they produce on a diverse pasture is more flavorful and more nutritionally complex than the milk produced on a single-species lawn.

Tom knows the names of probably forty plants growing on his farm, by sight, in the field. He learned most of them from his father, and his father from his father, and a few of them from the county Cooperative Extension agent who came out in the 1970s and taught Tom's father how to read pasture composition by walking a transect with a clipboard and counting hits. Tom can tell you that the upper pasture has too much fescue and not enough clover. He can tell you that the field by the road, the one with the seep at the lower edge, grows reed canarygrass that the cows do not love but will eat if it is what is there. He can tell you that two Aprils ago there was a hard frost on the eighteenth that knocked the alfalfa back by three weeks. He can tell you which paddock the

deer come through, where the woodchucks live, where the soil is shallowest over the bedrock and where it goes deep enough that you could push a fence post in by hand.

This is a kind of knowledge that nothing replaces and nothing transfers easily. It cannot be looked up. It is built one walk at a time, over decades, by a person who is paying attention to one piece of land. Most of what makes a real farmer a real farmer is contained in this kind of knowledge, and almost none of it is visible from the outside.

The cows will not stay in this paddock.

By tomorrow morning they will have grazed it down to three or four inches, and Tom will move them to the next one. The pasture is divided into about thirty-two paddocks of varying sizes, separated by a single strand of polywire that runs from a battery-powered fence charger and which Tom and Marcelino move every day. Each paddock gets a day or two of grazing and then a rest period of twenty-five to forty-five days, depending on how fast the grass is growing. In May, when the grass is exploding, the rest period is shorter. In August, when the heat slows growth, it is longer.

This is called rotational grazing, or sometimes managed intensive grazing, and the principle behind it is the partnership the grass and the grazer worked out long before us. Grass species — the wild ones, on the great prairies of North America before the cattle came — evolved alongside bison, who would graze a patch hard and then move on and not return for months. Grass that is grazed and then rested grows back vigorously. Grass that is continuously grazed, day after day in the same field, never gets a chance to rebuild its roots and eventually weakens. The old-

fashioned American practice — turning cows out into one big field for the whole summer — degrades pasture over time. Rotational grazing improves it.

This is not new science. The Zimbabwean ecologist Allan Savory has been writing and teaching about it since the 1960s. Some of his larger claims about using grazing to reverse desertification have not held up under careful study, and his work is contested in the academic literature. But the core idea — that mimicking the natural pattern of intense grazing followed by long rest builds healthy grassland — is well-supported. It is now standard practice on grass-based farms across the country, including Tom's. He moves a fence every morning. The whole system runs on a battery the size of a car battery and a few miles of wire and the knowledge, in his head, of how each paddock is doing this week and what it will look like in ten days.

So now: when you go to Whole Foods or the co-op or the grocery store, and you pick up a carton of milk, and the label says *grass-fed* — what does that mean?

It depends.

The USDA had an official Grass Fed label until 2016, when they withdrew it, because it was being applied so loosely that it had stopped meaning anything specific. After 2016, *grass-fed* became a marketing claim with no federal standard. Anyone who wants to can put the words on a label.

There are still real grass-fed certifications, run by private organizations. The strictest is the American Grassfed Association, which requires that animals eat only grass and forage their entire lives — no grain, ever, not even at the end — that they live on

pasture year-round when conditions allow, and that they never receive antibiotics or hormones. A carton with the AGA seal means something specific.

A carton that says *grass-fed* without that seal, or without some equivalent third-party certification, could mean almost anything. It could mean the cows were on grass for a few months in the summer and on a grain ration the rest of the year. It could mean the herd ate pasture grass during the day and a substantial scoop of grain at every milking. It could mean the cows were inside on a total mixed ration for most of their lives and the words went on the label anyway. The shopper has no way to know.

USDA Organic is a different certification, with its own rules. Organic does not require all-grass feeding, but it does require that organic dairy cows receive at least thirty percent of their feed from pasture during the grazing season, which has to be at least 120 days long. Organic also prohibits antibiotics, synthetic pesticides, GMO feed, and growth hormones. The seal means something. It does not mean grass-fed.

Tom's farm is both. The cows eat grass and forage and nothing else — fresh in the summer, baleage in the winter, never any grain — and the farm is certified organic and certified by the American Grassfed Association. The milk that leaves the farm carries both seals. It is, by the strictest definition the country has, the real thing.

It is also more expensive to produce than the milk on the shelf next to it. The cows produce less. The pastures cost more to maintain. The labor required is higher. We will come back to all of this.

For now: the cows are out.

By dusk the herd is still on the pasture. They have eaten about half of the available grass. Most of them have lain down. Some are still grazing. Some are chewing cud. The light is dropping toward the lake, and the grass is changing color in it, going from green to gold to a kind of amber gray. Hazel is at the lower end of the pasture, near the line of trees, lying with her front legs folded under her, her udder cool against the grass, her eyes half-closed. She has not had a day like this in five months.

Tom watches from the gate for a while. Then he goes back to the barn for evening milking. The cows will come up on their own when the time comes; they know.

Behind him, a hundred yards down the slope, in the row of hutches along the south wall of the heifer barn, Mabel is asleep on the straw. She is six weeks old. Her body has rounded out from the long-legged thing it was at birth. Her coat is shedding the slick of newborn for the heavier brown-and-white of a calf. She has never seen a pasture. She will, in late August, when she is weaned and grouped with the other heifer calves and walked out for the first time. Until then, the pasture is something she does not yet know is hers.

The cows graze in the falling light. Bertha is somewhere in the middle of the herd, where she has been for seven Aprils. Hazel is at the edge. Above them, the first stars are coming out over the lake.

It is a good day.

CHAPTER 5

5

Chapter 05 The Morning



A double-six herringbone parlor has six places on each side, stepped up from the floor, with the cows facing in toward each other at a forty-five-degree angle. The pit is a trench down the middle, three feet below the cow platform, where the milker stands. From the pit you can see twelve cows at a time — six on your left, six on your right — and the row of twelve udders hanging at chest height. The light is on. The vacuum pump is humming. The first six cows are on the left side and the next six are coming in from the holding pen on the right.

It is four-fifty in the morning.

The first cow up on the left side this morning is a Holstein-Jersey cross with a white face and a small dark patch over her left eye. Her name is Mabel. The older Mabel — there is a younger one

now too, in a hutch behind the heifer barn, and Tom keeps meaning to remember to give one of them a different name and keeps not doing it. The older Mabel is eight, and she has been in the milking string for six years, and she walks into her stanchion and lowers her head into the headlock without prompting and starts on her grain.

Tom approaches her from the right side. He is wearing rubber boots, rubber waterproof bib overalls, a long-sleeved shirt under a fleece, latex gloves. The first thing he does is take a small stainless-steel cup with a black plastic top — a strip cup — and squeeze a few squirts of milk from each of Mabel's four teats into it, one teat at a time. He holds the cup up to the light. The black surface is there to make any abnormality visible: a clot, a streak of blood, a flake of pus. He is checking for mastitis. He is checking for it on every cow at every milking, every day, for his entire adult life. Mabel's milk this morning is clean.

He sets the cup down. He picks up a foaming applicator of pre-dip — iodine, a deep purple — and dips each of the four teats in it, in turn. Then he turns to the next cow.

The pre-dip needs thirty seconds. While it is working, Tom moves down the line. Cow two: strip, dip, move. Cow three: strip, dip, move. Cow four: same. By the time he is at cow six the iodine on cow one has done its work, and he comes back. He pulls a clean white towel from a stack and wipes the iodine off all four of Mabel's teats in long firm strokes — one towel per cow, never reused, never shared, into a hamper at the end of milking. The wiping is also stimulation: it tells the cow's body that something is about to happen.

Then he runs his hand briefly over Mabel's udder. The udder is warm and full and a little firm; this is what it should feel like. He gives it a light press with the heel of his hand. Inside the cow's body, the press, the wipe, the iodine, the routine itself, the smell of the parlor, the sound of the pump — all of these tell her brain to release a hormone called oxytocin, which travels in her bloodstream from her brain to her udder and tells the cells of her udder to push milk down into the teat cisterns. This is called milk letdown. It happens within sixty seconds of stimulation. Without it, the milking unit cannot do its job. With it, the milk flows.

Tom takes the cluster — four short flexible cups joined by a manifold — lifts it up under Mabel's belly, and one at a time places each cup over a teat. There is a small twist of the wrist for each. The vacuum grabs each cup onto each teat with a soft pressing pull. Cow one is now on the machine.

He is at cow two before the unit on cow one has begun pulsing.

What is the machine, exactly?

A milking machine imitates a calf. This is worth saying clearly: every part of the machine has its origin in something a nursing calf does to its mother, and the engineers who designed the first machines in the late nineteenth century studied calves nursing in order to figure out how to do it.

A calf does not just suck. A calf works the teat between its tongue and the roof of its mouth, in a wave that moves down the length of the teat from base to tip, drawing milk out by squeezing it down rather than just pulling on it. The squeeze is the key. A pure vacuum, applied to a teat without rhythm, would not move milk efficiently and would damage the tissue. The squeeze releases

pressure on the teat between pulls, and lets blood flow back through the small veins of the teat skin. Without the release, the teat would bruise.

A milking machine does this with two systems. The first is a continuous vacuum, set at about twelve inches of mercury, which is what you would feel as a steady pull on your finger if you put it in one of the inflations. The second is a pulsator, which is a small valve that switches the air pressure between the inflation liner — the rubber sleeve that touches the teat — and the surrounding shell, sixty times a minute. When the pulsator opens to vacuum, the liner collapses around the teat and squeezes it. When the pulsator opens to atmospheric pressure, the liner relaxes and the vacuum on the teat draws milk down. Squeeze. Release. Squeeze. Release. Sixty times a minute. The milk flows through tubing to a receiving jar and from there, through a glass pipeline, into the bulk tank in the next room.

When the flow drops below a certain rate — the cow is finished — sensors in the cluster trigger a small cylinder that pulls the cluster down and off the udder automatically. This is called automatic take-off. Tom's parlor has it, like most modern parlors. Older ones did not. Older milkers had to watch each cow and remove each unit by hand, which was the way the work was done for a hundred years.

If you stand in the pit while a cow is being milked you can hear the machine breathing: the high steady whoosh of the vacuum, and underneath it the chuff-chuff-chuff-chuff of the pulsator opening and closing. You can hear the milk hitting the inside of the receiving jar in long irregular bursts. You can hear the cow shifting her weight on the platform above you, and the small click

of her hoof on metal, and somewhere down the line another cow letting her urine fall in a long warm stream onto the rubber matting.

The sound of a working parlor is, once you have spent any time in one, a sound you carry around with you for the rest of your life.

By five-twenty Tom and Marcelino are on round three. They have done thirty-six cows. They have forty more to go.

The work is the same on every cow, and it is also not. Some cows step up onto the platform like they are doing you a favor. Some cows balk. Some cows, who have been milked twice a day every day for years, walk into their stanchion, place their head into the headlock without prompting, take their grain, and stand still for nine minutes without ever needing a hand on their flank. Some cows are kickers. Tom has a list, in his head, of which cows are kickers and which kick because something is wrong and which kick for no reason; you learn to step back from the udder of a known kicker before reaching in. Some cows step on your foot. Marcelino, in his five years on this farm and at the previous farm where he worked for two years before, has had three broken toes. They were all on his left foot, because that is the foot the cow you are working on can reach.

Marcelino works on the right side; Tom works on the left. They talk to the cows more than they talk to each other. *Easy now. Easy. Buena, vaca, buena. Stand. Stand. Good.* Half of it is in English and half in Spanish, and most of it could be in either, because the cows are not really listening to the words. They are listening to the voice, which is steady, which is familiar, which is the same voice that has come to them at the same hour for as many mornings as they have been alive.

Hazel comes in on round four. She is on the left side, third from the back. Her udder has filled in the eight hours since the evening milking and her teats are, just slightly, swollen. Tom strip-cups her. Her milk is clean. He pre-dips. He wipes. He massages. He attaches. The cluster goes on with the small soft thunk of vacuum catching, and her milk starts to flow down the line, and her body relaxes the way every cow's body relaxes when the pressure of a full udder is being lifted from her, and Tom is already on to the next cow.

She is milked in four minutes and forty seconds. The auto-take-off pulls the cluster off her with a faint hiss. Tom returns, post-dips each teat in a different iodine — purple, the post-milking dip, which forms a thin barrier on the teat end against bacteria for the next several hours — and releases the headlock. Hazel walks out the back of the parlor, down the return lane, toward the bunk for her morning grain.

She has given, this morning, about three and a half gallons of milk. She will give about the same again tonight. Over the course of a year, discounting the two months at the end when she is dry before her next calf is born, she will produce something on the order of two thousand gallons. The average American drinks about seventeen gallons of milk in a year. Hazel feeds, by herself, about a hundred and twenty people.

The thing the work does to the people who do it is not visible at first.

Marcelino is twenty-nine. He looks twenty-nine. If you saw him in the parking lot of the Walmart in Geneva you would not look twice. But his hands, if you took one of them in yours, would tell you something. The skin on his palms is thick the way a

stonemason's is thick. The joints of his fingers are slightly larger than they should be at his age. His right shoulder is, very slightly, lower than his left. His knees ache in the morning.

Tom is fifty-six. His hands look older than that. There is a permanent stiffness in his right thumb that has been there since the early 2000s and which he attributes, accurately, to twenty years of attaching milking units. His back hurts where it has hurt since 1994. His left knee was replaced six years ago. He sleeps badly. He has been told by the only doctor he sees, who is in Penn Yan and whom he sees once a year because Sarah makes him, that his blood pressure is high, his cholesterol is high, and that he should consider what the doctor calls *slowing down*, which is a phrase that has no operational meaning on a dairy farm.

The hours alone would do this. The schedule alone would do this. Cows are milked twice a day, every day, including the day your daughter is born and the day your father dies and the day your roof collapses and Christmas. There is no version of the work in which any of these things give you a day off. If you do not milk the cows, the pressure builds painfully in their udders within hours, and within a couple of days you have mastitis running through the herd. Most dairy farmers, asked when they last took a vacation, will look at you the way you would look at someone asking when you last walked on the moon.

Add to the schedule: the bending. The lifting. The cold and the wet. The kicks and the steppings-on. The repetitive motion of attaching forty-some clusters in eighty minutes, twice a day, every day, for thirty years. The work does not break a person quickly. It

breaks them slowly. By the time they notice, they are halfway through being broken, and there is no exit they did not already foreclose by being who they are.

Tom would not say any of this out loud. Marcelino would not either. But it is true.

The cow's experience of the morning is harder to know.

She walks into the parlor on her own. She places her head in the stanchion on her own. She stands more or less still while she is milked. When the cluster comes off she walks out, on her own, and goes to the feed bunk. If she is one of the cows who likes the parlor, she will sometimes show up early in the holding pen and stand pressed up against the gate. If she is one of the cows who finds the parlor unpleasant, she will hang back, and Marcelino will have to walk down to her with a soft slap on the flank to get her moving, and she will walk in reluctantly, and stand reluctantly, and walk out as soon as she can.

What we can say with confidence: the milking unit, properly attached, does not hurt. The relief of a full udder being emptied is a real biological relief. Cows in a well-run parlor do not show signs of stress — no eye-whites, no rapid breathing, no stiffened bodies — and they show many signs of routine: chewing cud during milking, leaning against the stanchion, half-closing their eyes.

What we cannot say with confidence: what it is like to be a cow in a milking parlor at four-fifty in the morning, twice a day, for the entire seven or eight years of an adult dairy life. We do not have the language for it. We do not have the access. We have only what she does, and what she does is walk in.

By six-forty Tom and Marcelino have finished. The seventy-sixth cow is down the lane. The pit is empty. The vacuum pump is still running, because the wash cycle is about to begin.

The wash cycle is itself a small ceremony. Marcelino disconnects the receiving jar from the pipeline. Tom flips a series of valves in the milkhouse that send hot water with a chlorine-based sanitizer through every inch of pipe and tubing in the system, then a hot rinse, then an acid wash to dissolve milkstone, then a final rinse. The whole thing takes forty minutes and runs on its own. By the time it is done, the bulk tank in the milkhouse holds about two hundred and seventy gallons of milk at thirty-eight degrees Fahrenheit, and the parlor is clean enough to do surgery in, which it is not far from being.

The milk truck will come tomorrow morning at five. The driver, a man named Greg who has been picking up Tom's milk for fifteen years, will sample the tank, hook up his hose, and pump the milk into a tanker that already has milk from two other farms in it and will pick up four more on its route. The tanker will carry it to a processing plant in Cortland. From the plant, after pasteurization and packaging, some of it will go to the co-op's organic line. Some of it will go to a yogurt company in the Hudson Valley. Some of it will go, in cartons that say *organic grass-fed* and bear the seals the farm has earned, to grocery stores in New York and Connecticut and Massachusetts, where people will pick them up off the shelf without thinking about any of this and put them in carts and pay nine dollars and take them home.

But that is tomorrow. This is now.

At seven Tom comes back across the path — there is still snow in patches in April — and into the kitchen, where Sarah is at the stove and the radio is on. He has been awake for three and a half hours. He sits down. Sarah sets a plate in front of him.

He will be back in the barn by eight-thirty for morning chores: feeding calves, scraping alleys, pushing up feed, doing whatever the day asks of him next. By three-thirty in the afternoon he will be back in the pit. By six-thirty tonight he will have done all of this again. Tomorrow, at four in the morning, the alarm in the bedroom will go off, or it will not, because his body will wake him a few minutes before it does.

The work is the work.

CHAPTER 6

6

Chapter 06 June The Milk Check



The milk check from Tom's co-op arrives by direct deposit on the eighth of every month, for the milk that left the farm during the previous month. On the morning of June 8, 2025, Sarah is at the kitchen table with her laptop and a cup of coffee, and she is opening the bank's website to look at it.

The check for May is \$53,418.

She writes the number on a Post-it note, sticks it to the upper right corner of the laptop, and starts going through the deductions, line by line, on the co-op's online portal. The portal shows her the breakdown. The farm shipped, in May, 138,712 pounds of milk. That is 1,387 hundredweight — *cwt* in the language of the industry, *cee-double-yew-tee* the way it is said. The

price for that milk was \$38.50 per hundredweight. Times 1,387, that comes to \$53,400. Plus a quality bonus of \$263 because the herd's somatic cell count was below the threshold the co-op pays a premium for. Less hauling: \$554. Less the federal dairy promotion order, which is the assessment that pays for the *Got Milk?* ads and other things you and Tom have very strong opinions about: \$208. Less a few small administrative deductions.

Net: \$53,418.

This is what the farm earned for thirty-one days of work by two adults, one teenager when she's home, eighty-six dairy cows, and the soil of two hundred and ten acres in Schuyler County, New York.

It is more than most dairy farms in the country earned that month for the same effort.

It is not enough.

To understand why this is what dairy farmers get paid, you have to understand a system so old and so strange it would be hard to invent if you tried.

In the United States, since 1937, the price of milk has been set, in part, by the federal government. The system is called the Federal Milk Marketing Orders. There are eleven of them, each covering a region of the country. The Northeast Order covers New York and New England and parts of Pennsylvania and the mid-Atlantic. Tom's milk is priced under it.

The way the orders work is this. The government does not set one price for milk. It sets four, based on what the milk is going to be used for. Class I is fluid milk — the gallon you drink. Class II is soft products: yogurt, cottage cheese, ice cream. Class III is cheese.

Class IV is butter and milk powder. Class I gets the highest price. The other classes go down from there. The actual prices float month to month based on national market conditions; the formulas are public, and they would take a chapter of their own to explain, and have not made any farmer rich in seventy-five years.

Tom does not get to choose where his milk goes. His co-op pools the milk of all its members and sells it to processors all over the region. Some of it becomes drinking milk. Some of it becomes yogurt. Some of it becomes cheese. The farmer is paid a *blend price* — a weighted average across the classes — that reflects how the co-op's milk got used during the month. There is a great deal of math involved, performed by accountants in offices Tom has never been to, and the price comes out the other end.

What the farmer cannot do, under any version of this system, is negotiate his own price. He can switch co-ops, but the prices are similar. He can find a niche buyer — a cheesemaker, a yogurt maker — but the volumes are small and they have prices too. The dairy farmer is, in the language of economics, a *price-taker*: he produces a commodity, and the market gives him what the market gives him, and that is the deal.

The blend price for conventional milk in the Northeast in May 2025 was about \$22 per hundredweight. Tom is being paid almost twice that — \$38.50 — because his milk is certified organic and certified grass-fed by the American Grassfed Association. His co-op pays a premium for it, which covers the extra cost of producing milk this way. Without that premium, his farm would not be solvent. With it, his farm is, just barely.

A small grass-based organic dairy farm in Upstate New York has, broadly, the following economics in 2025.

The farm grosses about \$540,000 a year in milk income. That is the top-line number, and it sounds like a lot of money — most adults would think you were doing very well if your business grossed half a million a year. But this is gross.

Out of that \$540,000, in a typical year, Tom and Sarah pay:

- About \$60,000 in labor: Marcelino's wages, his housing, payroll taxes, workers' comp.
- About \$60,000 in costs related to the cows: the vet, AI semen, mineral supplements, hoof trimming, the organic and grass-fed certification fees, sawdust, the small daily things that add up over a year and ninety-four animals.
- About \$80,000 in equipment costs: the tractor, the skid steer, the milking system, the bulk tank, parts, fuel, depreciation on all of it.
- About \$90,000 in property and operating expenses: taxes on the land and buildings, farm insurance, utilities (the vacuum pump and the milk cooler run twenty-four hours a day), supplies, accounting, family health insurance — which is one of the largest single line items because Tom and Sarah are self-employed, and self-employed Americans pay full freight for health coverage.
- About \$50,000 in debt service on the loans they took out in 2014 to upgrade the milking parlor and in 2019 to buy the eighty-acre piece next door.

That comes to about \$340,000.

What is left — about \$200,000 — is what Tom and Sarah live on, pay taxes on, and reinvest into the farm.

Of that \$200,000, in 2024, they paid about \$50,000 in federal income tax, state income tax, and self-employment tax. They paid themselves about \$80,000 in family living expenses — for two people working roughly a hundred and twenty hours a week between them, plus a daughter when she is home, plus the value of being able to live on the land and not pay rent. The remaining \$70,000 went to reinvestment in the farm: a new water heater, a partial roof repair on the heifer barn, replacement of one of the milking inflations that had started failing, a small reserve toward the next thing that breaks, which something always does.

That is in a normal year.

In 2022, when the drought hit and they had to buy in baleage at twice the usual price, the family's take-home was \$34,000. They drew on savings to make the year work. In 2023, when one of the bulk tank refrigeration units failed and the repair was \$14,000 they had not budgeted for, the family's take-home was \$51,000.

This is what Tom and Sarah work for. This is the answer to the question, *what does a small organic dairy farm in the Finger Lakes earn?*

If you go to a Whole Foods in suburban Boston or Westchester or Greenwich, and you walk to the dairy case, and you pick up a half-gallon of organic grass-fed milk — the kind of milk that came from a farm like Tom's — you will pay somewhere between four and five dollars. If you buy a full gallon, somewhere between seven and nine. The exact number depends on the brand, the city, the year.

What you are paying for, and where the money goes, breaks down roughly like this. Numbers vary by product and region; treat them as the shape of it, not the precise figures.

For every nine dollars you spend on a gallon of organic grass-fed milk:

- About three dollars goes to the retailer for the cost of stocking, refrigerating, and selling it.
- About a dollar goes to the distributor who got the milk from the processor to the store.
- About two dollars and fifty cents goes to the processor who pasteurized the milk, packaged it in the carton, marketed the brand, and shipped it.
- About thirty cents goes to the truck that hauled the raw milk from the farm to the processor.
- About two dollars and twenty cents goes to the farmer who actually produced the milk.

Out of nine dollars, the farmer who got up at four-thirty in the morning every day for the entire month gets a little more than two dollars per gallon. Of that, after his expenses, he might keep thirty or forty cents.

This is not a moral judgment about Whole Foods, or about the processor, or about anyone in particular. The retail markup pays real costs. The processor's costs are real. The trucking is real. Everyone in the chain is, mostly, doing what their part of the system is set up to do. It is just useful to know what the gallon represents. It represents about thirty-five seconds of Tom's time, twelve hours of Hazel's, and roughly two cents per acre per year of the soil Tom's grandfather bought in 1947.

In 2003, there were a little over seventy thousand dairy farms in the United States. In 2023, there were about twenty-four thousand. In twenty years, two-thirds of the country's dairy farms went out of business. The trend has not slowed.

The farms that closed were, almost without exception, small. The farms that grew were large. The number of cows in the country is roughly the same as it was in 2003, because the surviving farms got bigger to make up for the ones that left. There are now dairy farms in California and Idaho and Texas with five and ten and fifteen thousand cows. There are dairies in New Mexico that are larger than some New England states' entire dairy industry. The American milk supply has not shrunk. It has consolidated.

Why? Costs go up. Milk prices fluctuate. Big farms have lower per-unit costs because they buy feed in bulk, run equipment more efficiently, and spread their fixed costs over more milk. Small farms cannot match these efficiencies. So in any given year, when prices drop or input costs spike, it is the small farms whose margins go to zero first. They sell. Their cows go to the auction. Their land gets bought by a neighbor or a developer. Their barns, in some cases, fall in over the course of a few winters.

The Hartleys have a neighbor, three farms over, named Jerry Schoonmaker. Jerry milked sixty cows on a hundred acres until 2018, when his knees gave out and his son did not want to come back from his job in Buffalo to take over, and he sold the cows at the Hosking auction in New Berlin in November of that year. The barn is still there. It is not in good shape. The land is rented to a

corn-and-soy operation owned by an investor in Albany whom Jerry has met once. Jerry lives in the house. He waves to Tom on the road.

This is part of what *organic grass-fed* costs, and it is part of what it is worth, and it is also part of what it cannot stop.

By the time Sarah is done with the spreadsheet it is nine-thirty. The May 2025 numbers are entered. The trailing-twelve-months tab is updated. The cash flow projection for the next quarter is on the next sheet over. She closes the laptop.

Tom comes in from the barn for coffee. He sees the Post-it note on the corner of the laptop. He does not need to ask.

"How was last month?"

"It was last month."

This is the same exchange they had in February, and in November of the year before, and in most months of the eleven years Sarah has been doing the books, because there are some answers that do not benefit from being more specific than they are.

Outside, in the lower pasture, Hazel is grazing.

CHAPTER 7

7

Chapter 07 August Haying And The People Who



Marcelino Pérez is twenty-nine years old. His full name, on his Guatemalan birth certificate, is Marcelino Mateo Pérez Hernández. He is, on this Tuesday afternoon in August, in the seat of a 1998 John Deere 6300 tractor in Tom Hartley's upper field, raking second-cutting hay into windrows under a sky that has not produced any cloud cover all day.

The temperature is ninety-one degrees. The cab on the 6300 has air conditioning that worked until 2017 and has not worked since. The window is open. Marcelino is wearing jeans, a long-sleeved cotton work shirt to keep the sun off his arms, scuffed leather boots, and a baseball cap that says CAYUGA TRACTOR & EQUIPMENT on the front. Sweat is running into the small of his

back and being absorbed by his belt. He has been raking hay for three hours, and he will be raking hay for another two, and then he will stop and eat something and then he will go and milk the cows.

The hay was cut yesterday by Tom on the discbine. The discbine is a machine that cuts the standing grass and lays it in long flat bands across the field, where it will dry in the sun for a day or two depending on the heat. Today, in this heat, one is enough. Tomorrow morning Tom will run the baler over the windrows and roll the hay into round bales the size of a small car, weighing about a thousand pounds each, and Marcelino will move them to the barn with the skid steer and stack them into the long row that will, eight months from now, in February, be the food the cows are eating off the bunk.

It is the most directly traceable arc on the farm: this grass becomes that meal becomes the milk that the truck takes to Cortland. The arc from this field to a glass of milk in a child's hand in the suburbs of Boston runs through, as much as anything else, the seat of this 1998 John Deere with its broken air conditioning, and the man who is in it.

Marcelino grew up in a town in the highlands of Guatemala called San Pedro Necta, in the department of Huehuetenango. The town sits at about five thousand feet and is built on the side of a steep slope above a valley, the kind of slope on which growing anything is hard and on which much is grown anyway: corn, beans, coffee, vegetables for the local market. His first language is Mam, an Indigenous Maya language spoken by about half a

million people in the western highlands of Guatemala. He learned Spanish in school. He learned what English he has from work, in five years of conversations in barns.

He left Guatemala in 2020. He was twenty-four. His daughter, Marisol, was two. He paid a coyote four thousand dollars, money he had borrowed from an uncle in Quetzaltenango, and he traveled overland through Mexico in a series of buses and trucks and one period on foot, and he crossed the Rio Grande at night near Roma, Texas, in a group of seven people with a guide. He was in U.S. immigration custody for three days and was released with a court date in Houston, which he did not appear for, because he had a cousin in upstate New York who had a connection on a dairy farm in Wayne County, and that is where he went. He has not been back to Guatemala since.

He has been on Tom's farm for five years. Before that he was on the farm in Wayne County for two. He has worked on a dairy farm every day of his life in the United States.

His wife, Lucía, lives in the same house Marcelino grew up in, with his mother and with Marisol, who is now seven years old. He sends home, on average, a thousand dollars a month. The money pays for the house, for Marisol's school, for the medicine when his mother needs it (she has diabetes), and for one ongoing project, which is the slow construction of a second floor on the house — concrete blocks, rebar, a tin roof — that Marcelino believes he will live in when he goes home, sometime in the next two or three years.

He has not held his daughter in five years.

About half of all the labor on American dairy farms is performed by foreign-born workers. The exact figure depends on how you count and what study you read; the most careful estimates put it between forty and fifty-five percent. The vast majority of those workers are from Mexico and Central America. The vast majority are men. A substantial proportion — most studies suggest seventy percent or more — are undocumented.

This is not a marginal feature of the American dairy industry. It is a structural one. If, tomorrow, every undocumented worker on every American dairy farm went home, the industry would not function within a week. Cows would not be milked. Calves would not be fed. Milk would stop arriving in trucks at processing plants, and yogurt cups and cheese sticks and gallon jugs would stop appearing on grocery shelves shortly after that.

Everyone in the dairy industry knows this. The farmers know it. The processors know it. The retailers know it. Congress knows it. The agency that enforces immigration law knows it. There is no version of the American milk supply, in the year 2025, that does not depend on this labor.

How did this come to be? Through a series of choices made over the last three decades, almost all of which involved someone deciding not to do something.

The agricultural guest-worker visa in the United States is called the H-2A visa. It allows foreign workers to come legally to the U.S. to perform farm work, with employer sponsorship, for a defined period of time. In 2024, about three hundred and fifty thousand H-2A visas were issued. Workers picked apples in Washington

and strawberries in California and tobacco in North Carolina and lettuce in Arizona, all legally, all under the protection of a federal program with rules about housing and wages and transportation.

None of those workers can be on a dairy farm. The H-2A visa is, by law, for seasonal work only. Dairy is not seasonal. Cows are milked every day of the year. A worker on a dairy farm cannot be hired under H-2A, because the work has no off-season, and the law has been written so that it does not apply.

The dairy industry has been lobbying Congress to fix this for thirty years. The fix is not technically difficult — a year-round agricultural visa, with appropriate protections, would solve it. The fix has not happened, because immigration is one of the most politically charged issues in American life, and any bill that addresses any part of it gets pulled into the larger fight, and the fight does not resolve. So the law has stayed roughly the way it was written in 1986. And dairy farms have hired the workers they could find, who are mostly undocumented, who do the work, who keep showing up.

Tom did not write this law. Marcelino did not choose this law. They both work inside it.

At twelve-thirty Marcelino comes back to the trailer for lunch.

The trailer is the same trailer Tom's father put in for a hired man in 1991 — a single-wide on cinder blocks behind the barn, with a propane tank visible from the kitchen window of the main house and a small wooden porch that Marcelino built himself in his second year. The inside is small but it is his. There is a couch, a television (Univisión, mostly soccer), a kitchen with an electric kettle and a four-burner stove, a bathroom, a bedroom in the back.

The walls have a few things on them. A small wooden cross over the bed. A photograph of Marisol at three, in a school uniform, smiling. A printed-out picture of the Guatemalan national soccer team. A calendar from the Spanish-language Mass at Saint Stephen's in Geneva.

For lunch he heats up beans and rice from the night before in a cast-iron pan, makes two tortillas on the burner, and sits at the small kitchen table to eat. He has done this almost every day of the last five years. He likes it. The food is the food his mother made and his wife makes and that he has made himself for so long that he has stopped thinking about it.

While he is eating, he calls home. The time difference between upstate New York and Guatemala is two hours; in San Pedro Necta it is ten-thirty in the morning, and Lucía is at the market. Marisol is at school. His mother is the one who answers.

Mijo. ¿Cómo estás?

Bien, mamá. Hace calor.

¿Cuánto?

Como noventa.

His mother makes a sound that means *hot*. They talk for ten minutes about nothing — the weather, the price of corn, the leak in the kitchen ceiling that Marcelino is paying to have fixed for the third year in a row, what his cousin in Quetzaltenango is doing now. He does not ask if she is taking her insulin. She always says yes. He worries about it anyway.

Avísale a Marisol que su papá la quiere.

Le avisaré.

He hangs up. He finishes the rice. He washes the pan. He goes back to the field.

Marcelino is paid sixteen dollars an hour. He works approximately seventy hours a week, sometimes more in haying season, sometimes less in winter. His annual earnings, before any deductions, are about fifty-eight thousand dollars. He is also provided with the trailer, the propane for heating it, water, and electricity, which together represent about another eight thousand dollars in annual value.

Tom pays him as a 1099 contractor, which is legally questionable for someone who is in fact an employee, and which is the way most undocumented dairy workers are paid because it does not require a Social Security number that matches the employer's records. Marcelino does not file a U.S. tax return. Tom does not withhold taxes. The arrangement is a fiction that both sides understand.

By the standards of American dairy labor, Marcelino is paid well. The median wage for a dairy worker in the United States in 2024 was about fourteen dollars an hour, with no housing. The bottom quartile is below twelve dollars. The hours are generally similar to what Marcelino works, which is to say long. Tom, for what it is worth, is one of the better employers in the region. He pays more than the median, provides housing he keeps in repair, pays Marcelino for the days he is sick, and has, twice in the last five years, offered him an unpaid two-week visit home. Marcelino did not take it either time. He is afraid that if he leaves, he cannot come back.

What does this kind of life cost a person?

The body shows it first. Marcelino's hands, at twenty-nine, look forty. His lower back hurts most mornings; his knees ache in the cold. He has not been to a dentist in five years. He has not been to a doctor in five years for anything other than a fever in 2022 that Sarah drove him to a clinic in Geneva for and paid for in cash, because Marcelino does not have insurance and could not have used it if he had.

The mind shows it differently. Marcelino is not a sad man. He is a determined one. He came here to do something, and he is doing it. But there are things in him that he carries: the absence of his daughter; the quiet awareness that she is growing up without him in the room; the question of whether the second floor of the house in San Pedro Necta will be ready in two years or three, and whether he will be ready when it is. He goes to Mass when he can. He prays. He sends home what he sends home. He sleeps.

The thing he says about it, when asked, is *Es lo que hay*. It is what there is.

On Sunday afternoons, Marcelino is mostly off. He milks in the morning, with Tom, the same as every other day. After breakfast, he is free until evening milking. About once a month, on a Sunday, Sarah drives him into Geneva.

The drive is twenty-two minutes. Marcelino sits in the passenger seat. Sarah talks about whatever Sarah is talking about — Anna's grades at Cornell, the price of beef at Wegmans, what is happening in the news. Marcelino listens. He nods. Occasionally he says something. Once, four years ago, he made a joke in English about Sarah's driving that made her laugh until she nearly went off the road, which he found to be against the joke's larger point and which he has never repeated.

At the Western Union counter, in the back of a check-cashing place on Exchange Street, Marcelino sends \$400 to Lucía. The fee is \$9.99. The money will be available to her in twelve hours, in the equivalent in Guatemalan quetzales. They have done this on the fifteenth of every month for sixty months. Lucía will pick it up at the bank in Huehuetenango on Monday.

Then Sarah drives him to Saint Stephen's, where the noon Spanish Mass is just letting out, and Marcelino goes in to confess, and Sarah goes around the corner to the Wegmans to do the week's grocery shopping. She picks him up in an hour. They drive back to the farm. He goes to his trailer. He calls home. Marisol is awake by then. She is seven. She tells him about a friend at school who has a new bicycle. He tells her he loves her. He says it many times in their conversations, because there is a thing in him that does not believe she has heard it enough.

In the evening he milks the cows.

A book about an American dairy farm in 2025 cannot be honest without saying these things. The milk in the carton is what the cow makes. The cow is what the grass and the weather and the genetics make. The work that turns the cow into the carton is, on a great many farms in America and on this farm too, what a man like Marcelino does, in a country that does not see him, for a family that has not seen him in five years, on a tractor whose air conditioning broke in 2017.

The farm Tom runs is a good farm. The wages Tom pays are better than most. The trailer Marcelino lives in is dry and warm and has good light. None of this changes what the system is: a system no one designed to be good. It is a system that grew up in

the gap between two sets of laws — one that says we need this labor, one that says this labor cannot legally exist — and it has been allowed to remain in that gap for thirty years.

Marcelino Pérez is twenty-nine years old. His daughter Marisol is seven. The last time he held her, she was two. He sends her four hundred dollars on the fifteenth of every month, and he calls her on a phone he keeps on speaker so she can hear what is happening even when there is nothing to say.

Outside the trailer, where Marcelino is asleep, the August night is loud with crickets. Down the slope, in the main barn, the cows are settled in their stalls. In the upper field, the windrows of hay are drying in the dark. Tomorrow Tom will bale them. Tomorrow at four in the morning, Marcelino will get up and walk to the parlor.

It is the work he came here to do. It is not all of him.

CHAPTER 8

8

Chapter 08 Goats Sheep And The Cheese Acros



In eastern France, in the mountains of the Jura along the Swiss border, a cheese called Comté is made under a set of rules that have been in place since 1958. The rules are not, primarily, about taste. They are about who gets to make a living.

The cheese itself is excellent. It is a hard cheese, aged anywhere from four months to three years, with a flavor that ranges from fresh and grassy at the young end to nutty and almost caramelized at the old end. About seventy thousand tons of it are made every year, more than any other appellation cheese in France. If you have eaten Comté in the United States — at a cheese counter, in a restaurant — you know what it tastes like. What you may not know is what it represents.

Comté is made under what the French call an *appellation d'origine contrôlée*, an A.O.C., which is a legal designation that says: this cheese can be called by this name only if it is made in this place, by these methods, from milk produced under these conditions. The A.O.C. for Comté is roughly forty pages of rules. Among them:

- The cheese can only be made in a defined geographic zone covering parts of three French departments — the Jura, the Doubs, and the Ain — totaling about three and a half million acres.
- The milk can only come from two breeds of cow, the Montbéliarde and the French Simmental, both adapted to the Jura's mountain pastures.
- The cows must spend at least half the year on grass. They cannot be fed silage of any kind.
- The land must be stocked at no more than approximately one cow per hectare — about two and a half acres per cow, which is extraordinarily low and means that the pastures are not pushed.
- The milk cannot be transported more than twenty-five kilometers from the cow to the cheesemaker. About fifteen miles.
- The cheese must be made in copper vats, by hand, in cooperatives the French call *fruitières*.
- The aging must be done in caves at controlled humidity and temperature, for a minimum of four months and usually much longer.

About three thousand dairy farmers, with herds averaging maybe forty cows each, ship their milk to about a hundred and fifty *fruitières* scattered across the region. The *fruitière* in any given village is owned, cooperatively, by the farmers who supply it. The cheesemaker is an employee of the cooperative. The cheese is then sold, mostly to specialized aging houses called *affineurs*, who hold it in their caves for the necessary months and then sell it on. Profits flow back through the chain, and a substantial share — much more than in the American system — ends up in the hands of the farmers who produced the milk.

This is the thing worth being clear about. A Comté farmer with thirty cows can make a living. A Comté farmer with thirty cows can pay her workers, raise her children, send those children to university, retire, and pass the farm on. A New York dairy farmer with thirty cows cannot.

How can this be?

Several things at once.

The first is the appellation itself. Because Comté can only be made in a specific way in a specific place, no industrial dairy in another part of France can flood the market with Comté-labeled cheese. The supply is, in effect, capped. Demand has grown over decades. Price has risen with demand. The premium farmers receive for milk that goes to Comté is roughly thirty to forty percent higher than the price for non-appellation milk. Over a year, on a small farm, that premium is the difference between solvency and not.

The second is the cooperative structure. In the American dairy system, as we saw in June, the farmer sells milk to a co-op, the co-op sells it to a processor, the processor sells the finished product, and most of the markup at each stage stays at that stage. In the Comté system the farmer is part of the entity that makes the cheese. The *fruitière* is owned by its members. When the cheese sells, the value flows back. The cheesemaker is paid a salary, the *affineur* takes their cut, but a large share of the residual goes to the people who produced the milk.

The third is the cultural and political agreement that this is worth doing. The French state has, since the early twentieth century, maintained a policy of agricultural protection that is foreign to most American thinking about food. The A.O.C. system covers more than three hundred French products, from wines to cheeses to meats to lentils, and it is administered by a federal agency with substantial enforcement power. The French government and the European Union both subsidize small farmers in ways the American government largely does not. The point of all of this is not, primarily, economic efficiency — bigger industrial farms could produce more food more cheaply. The point is the preservation of a way of life, a tradition of food, a rural population, a landscape. The French have decided, through their politics, that these things are worth paying for.

Americans have, broadly, decided otherwise.

Within the United States, where no equivalent system exists, some small dairy farmers have made a different choice. They have raised goats or sheep instead of cows.

A dairy goat weighs about a hundred and fifty pounds, an order of magnitude smaller than a cow. A dairy goat gives one to two gallons of milk a day, depending on breed and management — perhaps a fifth of what a cow gives. The milk is high in butterfat, moderately easy to digest, and has the slight tangy quality you taste in good *chèvre*. Goats are also browsers — they prefer brushy plants, vines, leaves — and they can use land that a cow cannot use, which on hilly New York property turns out to matter.

A dairy sheep is similar in size to a goat, about a hundred and fifty pounds, but gives much less milk: perhaps half a gallon a day at peak, much less than a goat or a cow. The milk is rich. The fat globules are smaller than in cow milk, and the milk itself contains roughly twice as much fat and protein per gallon. It is the milk that makes *Roquefort* in France, *Pecorino* in Italy, *Manchego* in Spain.

The economics of small ruminant dairying are different from those of cow dairying in important ways. The animals cost less. The barns cost less. The milking equipment costs less. The feed required is less. The land required is less. A small farmer with twenty acres and limited capital can plausibly start a dairy operation with goats or sheep. The same farmer, on the same land, would have a very hard time starting a cow dairy.

The trade-off is the market. Goat and sheep dairy products are niche in the United States. Almost all of the goat milk produced is made into cheese, sold either directly to consumers at farmers markets or to specialty wholesalers. Sheep dairy is even smaller.

There are perhaps a thousand commercial sheep dairies in the entire country. The total American sheep dairy industry is smaller, in volume, than a single large cow dairy in California.

But for some farmers the smaller market is a feature, not a bug. The price per gallon for goat or sheep milk, sold as cheese, is far higher than the price per gallon for cow milk. The customer base is people willing to pay more for an artisanal product. The work is hard, but the scale is human. A small goat dairy with thirty animals can be run by one or two people, with much less equipment and much less labor and much less capital than a small cow dairy. The path to viability is harder to find, but easier to walk once found.

There are not many of these farms. Most that are tried do not last. The ones that do tend to share certain features: the operator already has another source of income (a spouse with a job, a partial pension, a remote tech career); the farm produces a value-added product (cheese, not raw milk) that the farmer is making and selling directly; the farm has built a local customer base over several years; the farm has accepted that the operator's income will be modest for a long time, possibly forever.

Tom does not raise goats or sheep. He has thought about it. He has friends who have done it. They have, on the whole, smaller herds, smaller farms, smaller checks at the end of the month, and slightly less back pain. They are, on average, holding on.

What is the use of telling Americans about Comté?

It is not, primarily, to argue that the United States should adopt the A.O.C. system. The political and cultural conditions that produced the French system over two centuries are not present in

the United States, and a top-down attempt to copy it would, in all likelihood, not work. The American food economy is different — bigger, more industrial, more tied to commodity markets, less anchored in regional tradition. To say *we should be more like France* is the kind of thing that sounds good at a dinner party and is hard to operationalize.

The use of telling Americans about Comté is the same as the use of telling them about anything from elsewhere: it makes visible the assumptions inside the system they live in. An American who has never heard of A.O.C. tends to assume that the way American dairy works is more or less the way dairy works. It is not. There are systems in which a thirty-cow farmer can make a living. There are systems in which the price of milk is set partly by tradition, partly by region, partly by the deliberate decision of a society to keep small producers in business. There are systems in which the rules around cheesemaking are written in cooperation with the cheesemakers themselves, rather than handed down by a federal agency a thousand miles away. These systems exist. They are working today. They are real, not theoretical.

Whether any of this could ever apply to Tom's farm in Schuyler County is an open question, and probably not the right one. The right question is the one a reader of this book can hold for a long time, without rushing to answer: *Why is the system the way it is, and what would it take to make it different?*

Tom and Sarah have eaten Comté. Sarah brings it home from Wegmans every December for the Christmas cheeseboard. They know what it is. They know it is made by small farmers in the

mountains of France. They have, once or twice over the years, joked between themselves that they were born on the wrong continent.

But Tom is a New York dairy farmer in a system that is American, and his farm is the farm he has, and he is the farmer he is. He produces, every year, about a hundred and fifty thousand gallons of milk on two hundred and ten acres of land, with the help of one full-time worker, his wife when she has time, his daughter when she is home, and ninety-some cows. He sells the milk through his co-op to a processor who sends it to grocery stores. He gets a check that depends on prices set by other people in other rooms. The milk leaves the farm in a tanker. The check arrives by direct deposit on the eighth of the month.

There is no Comté in this. There is only this farm and these animals and this work and this checkbook. It is, in the language of the system Tom lives inside, a successful operation. It is also a place a person could lose, in any given year, depending on the weather, the prices, the breakdowns. It is, like every farm of its size in America, the survivor of a system that has, for the past three decades, been quietly removing the others.

The cheese in the suburbs is real. The farms that made it are real. The fact that those farms are not American farms is also real.

That is, for now, the thing to know.

CHAPTER 9

9

Chapter 09 October Breeding Back



It is the third week of October, and the leaves on the maples along the road are well into their orange and red, and Hazel is in heat.

That last part is something Tom has known for about an hour. She is at the back of the milking string, breathing in the steady focused way a cow breathes when something specific is happening inside her, but earlier this morning, in the parlor, she had the distinctive look of a cow about to come into heat, and the activity monitor on her collar had been showing higher-than-baseline movement since the previous evening, and at lunchtime in the dry

pen Tom had watched her stand still and let another cow place her front feet on Hazel's hindquarters for several seconds. Standing heat. The thing he had been waiting to see.

Hazel has been open since May.

That word — *open* — is what dairy farmers call a cow who is not pregnant. The opposite is *bred*. A bred cow is a cow whose body is, on the slow internal clock of nine and a third months, doing the work of building the next calf. An open cow is a cow whose body is between things. Open cows do not contribute, in dairy economics, what bred cows contribute. An open cow, the longer she stays open, becomes a problem.

Hazel was bred in late May, six weeks after Mabel was born. The semen was from a Jersey bull called 7JE08824, whose catalog photo shows him standing in profile in front of a fence, looking middle-distance, with the slightly bored expression of a celebrity at a press event. The breeding did not take. Tom checked her thirty days later and her uterus, by palpation, was empty.

He bred her again in mid-June. That did not take either.

By that point it was July, and the temperature was in the high eighties almost every day, and Tom did what every grass-based dairy farmer in the Northeast does when summer arrives: he stopped trying to breed his cows. The reason is heat. A cow in temperatures above seventy-five degrees, day after day, with a body that already runs warm and an internal furnace of forty pounds of milk a day, is in what dairy people call *heat stress*. Heat stress reduces conception rates by twenty to thirty percent. It also

lowers milk production, weakens immune function, and makes everything in a cow's life harder. You can breed her if you want; she will mostly not get pregnant. So you wait.

You wait until October. The leaves change. The temperature drops. The cows' bodies come back to themselves.

This morning, Hazel is in heat.

How does Tom know? Two ways. The collar and the eye.

The collar. Around Hazel's neck is a small black device — about the size of a deck of cards — that contains an accelerometer and a microcontroller and a battery. It tracks her movement. Cows in heat are restless: they walk more, they stand for shorter periods, they spend more time at the feed bunk and less time lying down. The collar collects this data and sends it, via a base station in the milkhouse, to a piece of farm management software Tom looks at every morning on his phone. When a cow's activity spikes above her own baseline, the software sends Tom an alert. Hazel's alert came in at six-thirty last night.

The collar is a relatively recent addition to the farm — Tom installed the system in 2019 — and it has roughly tripled the rate at which his herd's heats are caught. Before the collars, dairy farmers detected heats by walking through the herd several times a day and looking. Looking still works. The collar just makes it less of a coin flip.

The eye. Even with the collar, Tom watches. He watched at five this morning when he brought Hazel in from the pasture. He watched at eleven when he was pushing up feed. He watched at noon when he saw the other cow place her feet on Hazel's hindquarters. The collar said *Hazel is showing increased activity*

consistent with estrus. The other cow said *Hazel is in heat now*. The two sources of evidence add up. Tom is going to breed her this afternoon.

Why this afternoon? The biology of timing.

A cow's estrus cycle is twenty-one days long. For most of that cycle her ovaries are doing slow internal work that produces no outward sign. For about twelve to eighteen hours of every cycle she is in *standing heat* — the brief window when her body is releasing an egg and is biologically ready to be bred. That window is the thing Tom is trying to catch.

The egg, once released, lives for about ten hours in the cow's reproductive tract before it deteriorates. The sperm, once introduced, lives for about twenty-four hours and is most vigorous in its first six. The two have to meet in a specific section of the oviduct at a specific time. If you breed her too early — before standing heat starts — the sperm may die before the egg arrives. If you breed her too late — more than ten hours after the egg has dropped — the egg may be gone by the time the sperm makes it. The optimal window for AI is somewhere in the middle of standing heat, give or take a few hours. Tom watched her at noon. She is most likely in the middle of her standing heat right now. The window of greatest fertility closes by tonight.

So this afternoon, between three and four, while Marcelino starts the evening prep in the parlor, Tom will go to the AI tank in the milkhouse, take out the small straw of frozen semen he selected three weeks ago, thaw it carefully, load it into a stainless-steel insemination gun, walk back to the cow he has known for six years, and put a calf inside her.

The semen comes in a thin plastic straw about the size of a coffee stirrer. Each straw holds about a quarter milliliter of bull semen — roughly fifteen million sperm — frozen in liquid nitrogen at minus three hundred and twenty degrees Fahrenheit. The semen is stored in a tank in the milkhouse, a tank Tom keeps filled with liquid nitrogen, which is replenished by a delivery service from Ithaca every three weeks.

The semen Tom selected for Hazel this round is from a Jersey bull called 7JE16204 — different from the one that did not take in May. Tom used to breed her to Holsteins for the volume; he stopped a few years ago, when he made the math come out for himself: on grass, his Jersey-cross daughters held up longer and cost less to keep, even at lower production. Mabel, who is upstairs in the heifer barn now, is the first of Hazel's calves from this new arithmetic. The catalog from Select Sires, which is one of the largest of the dairy genetics companies in the country, has a page on this bull. His name is Greenview Dior. He was born in 2019 on a farm in central Wisconsin. He is a Jersey of strong grass-based parentage. His daughters, the catalog says, calve easily, produce well on grass, and have what the genetics people call *good udder conformation*, which means their udders attach to their bodies in ways that hold up over time. A straw of his semen costs Tom thirty-two dollars.

This is one of the strange things about modern dairy.

A bull at a place like Select Sires lives in a barn somewhere in Ohio or Wisconsin and is collected, by trained technicians, two to three times a week. His semen is processed, evaluated, divided into straws, and frozen. From any one collection the company may produce three hundred straws. Over the course of his productive

life, a single popular bull may sire — through artificial insemination — fifty thousand or one hundred thousand or three hundred thousand calves on dairy farms across the United States and around the world. He never meets a single one of them. The cows never meet him. The semen travels in liquid-nitrogen tanks in the back of refrigerated trucks; the bull stays where he is until he dies or stops being productive.

A modern dairy cow's father is, almost invariably, a bull she will never see. So is her grandfather. So is, going back many generations, the entire male line of her family tree. A small number of bulls have, in this way, become the genetic ancestors of an enormous portion of the world's dairy cattle. Mabel — the Mabel born in March, who is now seven months old in the heifer barn — has, in her ancestry, perhaps two dozen bulls in the past century who account for a majority of her DNA.

This is genetic management at industrial scale. It is also not, in any meaningful sense, what the word *breeding* used to mean.

At three-fifteen Tom takes the semen straw out of the tank with a long pair of tweezers, drops it into a small metal cup of warm water at thirty-five degrees Celsius for forty-five seconds, dries it, slides it into the insemination gun — a stainless-steel rod about the length of his forearm — and walks back through the milkhouse to where Hazel is now standing in a head-lock at the end of the milking string, contained but not stressed, eating a small handful of grain.

He puts a long plastic glove on his right arm — the glove goes up to his shoulder — and reaches into Hazel's rectum with his gloved hand. Through the wall of her rectum he can feel her cervix and her uterus, and he uses his hand from inside her body to

position her cervix in line with the gun. With his other hand, the bare one, he inserts the insemination gun through her vulva, through her vagina, finds the cervix with the tip of the gun, and threads the gun through the soft folds of the cervix into the body of her uterus. There are five or six small folds in the cervix, and the gun has to navigate each of them. This is the part that takes practice. Tom has done this approximately four thousand times in his life. He is good at it.

When the tip of the gun is in the right place — at the junction of the uterine horns, where the egg will be coming down — he presses the plunger. The semen is released. He withdraws the gun. He withdraws his arm from her rectum. He removes the glove and drops it in a hamper. The whole procedure has taken about three minutes.

Hazel, throughout, has been chewing her grain. She has, as far as one can tell, noticed almost nothing.

Why are we doing all of this?

The biological answer: a cow gives milk because she has had a calf. The system does not work otherwise. Lactation, in mammals, is part of the reproductive cycle. The udder fills up because there is a calf to nurse, and the udder will keep producing milk as long as it keeps getting emptied — by the calf, or, in dairy, by the milking machine — but not forever. Milk production peaks about six to eight weeks after calving and then declines, slowly at first and then more steeply, until at about ten months the cow is producing too little to be worth milking anymore. To keep her in production beyond that, you have to start the cycle again. You have to breed her back.

The economic answer: every day a cow is open is a day she is moving toward the end of her productive lactation without anything to follow it. If she does not get bred back, eventually she runs out. She has to be dried off, at some point, regardless. If she is pregnant when she is dried off, she will calve in two months and start a new lactation. If she is not pregnant when she is dried off, she is simply a cow standing in a barn eating feed and producing no milk, until you decide what to do with her. *What to do with her* is, on every dairy farm, an active question.

A cow who has been bred unsuccessfully four or five times, with no clear medical cause, is a cow whose value to the operation is approaching zero. There is a number, in every farmer's head, of dollars per day she is costing him: feed, bedding, water, parlor time, the opportunity cost of the stall she is in. If that cost runs long enough, she will be culled. She will be sent to the auction or to a beef farmer. Her place in the herd will go to a heifer who has just freshened — a young cow in her first lactation — and the cycle will continue.

This is what the cow's life is, in exact terms. She lives so long as she keeps producing. She produces because she is bred. She is bred because the system requires her, on a roughly annual cycle, to start a new pregnancy. The continuity of her life is contingent on the continuity of her pregnancies. Hazel's existence on this farm is the expression, in the body of one animal, of an arrangement that has been worked out over centuries between the cow and the human, and which is, by both biology and economics, what it is.

By four o'clock Hazel is back in the alley, walking slowly toward the water trough. She drinks. She turns her head and looks at Tom, who is in the milkhouse cleaning up the AI gun. She looks back at the trough. She drinks again.

In thirty days Tom will check her. If her uterus has the small firm shape of a six-week pregnancy, she has taken. If it does not, she will come back into heat in three weeks, and Tom will breed her again, and they will see.

The light through the milkhouse windows is yellow now, the long horizontal yellow of October afternoons in the Finger Lakes. Outside, in the heifer pasture, Mabel is grazing. She was weaned in August and turned out with the other heifers in early September. She is half her mother's height already. By next winter she will be the size of a small adult cow. By a year and a half from now she will be ready to breed. She does not know any of this.

Hazel drinks her water. Tom puts the gun away. The sun moves another inch across the floor of the milkhouse. Outside, somewhere over the hill, a tractor is running, and Marcelino is on it, and one of the things that gets done in October on a farm that does not stop is getting done.

CHAPTER 10

10

Chapter 10 November The Decision



It is the second week of November. The trees along the road are bare. There has not been a hard freeze yet, but it is coming. The grass in the lower pasture is yellowing at the tips. The cows have been brought in from the night pasture for breakfast and the morning milking is done. Tom is in the parlor, in a long plastic glove up to his shoulder, with his arm in Hazel.

This is a thirty-day pregnancy check.

For a few seconds his face is the face of a person doing two things at once. He is feeling, through the wall of her rectum, for a particular shape and size in her uterus — the small firm fluctuation of a six-week pregnancy, which feels, to a hand that has felt thousands of them, like nothing else. He is also, although he

would not admit this to anyone, hoping. Hazel is six. She has been on the farm her whole life. She is in her sixth lactation. She has had five calves and one of them is named Mabel and is in the heifer barn behind him. She has been open since May, which is too long, and if she has not taken this time he is going to be in a conversation with himself this weekend that he does not want to be in.

He moves his hand. He moves it again.

She is pregnant.

He says nothing — she would not understand him if he did, and there is no one else in the parlor to hear it — but he leans his forehead, briefly, against her flank, and breathes out. Then he straightens up. He withdraws his arm. He removes the glove and drops it in the hamper. He writes *Hazel #47, +30* in the herd book in pencil. She will calve, by his calculation, in the second week of August, when the corn across the road is tall and the air is heavy and the second cutting of hay is being put up.

But this morning, before any of that, Tom has a different thing to do.

This morning, Tom has a list. The list has three names on it. The names are Bertha, Pearl, and Daisy. They are in a yellow legal pad in the top drawer of the desk in the milkhouse. They were written down last night, after dinner, by Tom and Sarah at the kitchen table, with the herd records open on the laptop and a glass of bourbon between them, neither of them saying very much, both of them having had this conversation many times before.

Today is the day Tom and Sarah cull.

Culling is the dairy industry's word for the process by which a cow leaves a milking herd. It can mean a number of different things. In the language of the farm, a culled cow is a cow whose time on this farm is over. She is sold — to a beef operation, to an auction, occasionally to another dairy that wants her — and her stall in the barn is filled by the next heifer who has come due.

About one in three cows in any given American dairy herd is culled each year. The figure has been remarkably stable for decades. On a herd Tom's size, that means about thirty cows leave every year. Some leave for production reasons (their milk has dropped below what makes them worth keeping). Some leave for reproduction reasons (they have failed to get pregnant after enough services that the math no longer works). Some leave for health reasons (chronic mastitis, lameness, metabolic disease). A few leave because they have, in the language farmers use, *gone bad* — become aggressive, become kickers, become unmanageable in the parlor — although this is rare on a small farm where people know each cow.

Most of the cows that leave do not leave to retire. American dairy cattle, when culled, almost always go into the beef supply chain. Their meat — leaner and tougher than meat from purpose-raised beef cattle — becomes ground beef. It becomes hamburger. It becomes the patties that anchor most of the meat consumption of the United States. The single largest source of beef in the American food supply is, statistically, dairy cows at the end of their dairy careers.

This is part of dairy. It is part of why dairy and beef are not separate industries; they are two parts of the same industry, with the same animal at different points in her life. The grass-fed

organic milk in the carton at Whole Foods, and the ground beef in the same shopping trip, are likely to have come from the same kind of cow, raised on the same kind of farm, processed at the same kind of plant.

Bertha is nine years old. She is the old Brown Swiss who walks out of the barn first on turnout day every spring and goes immediately to the grass without looking up. She has had eight calves. She has been the steadiest milker in the herd for most of those years, and she has never been a problem for anyone. But this fall her production is down. Her milk in May was around fifty-five pounds a day. Her milk in October was thirty. Her udder, when Tom presses it, is softer than it should be at peak lactation; there has been a small ongoing infection that the standard organic treatments have not cleared, and which Tom does not want to put antibiotics into because the farm is organic and antibiotic use complicates everything. Bertha is a cow who has done her work, and the work is done.

Pearl is five. She has been bred eight times in the past sixteen months, and not one of those services has resulted in a confirmed pregnancy. The vet has examined her twice and found no obvious medical cause. Pearl is, in the language of the farm, *infertile in a way that doesn't have a name*. She is going dry in two weeks regardless. After that she would be a cow standing in the barn eating feed and producing nothing, indefinitely, while the dollars-per-day arithmetic ran in the wrong direction. There is nothing wrong with Pearl exactly. She is just not a cow the system can afford to keep.

Daisy is four. She is in her second lactation. She has had three episodes of mastitis in the past nine months, two of them in the same quarter — the same teat — which is a sign of a chronic infection that has likely entered the udder tissue and may not respond to anything. The veterinarian's recommendation last month was to cull. Tom has known this. He has been waiting for the right week, which is now, because the truck for the auction in New Berlin runs on Wednesdays and today is Tuesday.

These are the three. There are probably four other cows in the herd who are heading toward the same decision in the next two months. Tom and Sarah will go through this exercise again before Christmas, and again in February. Thirty cows a year. About one every twelve days.

The arithmetic of the decision is harder than anyone outside the farm thinks it is, and easier than most farmers admit.

It is harder than it looks because it is not just one number. It is a vector of numbers — production, conformation, fertility, health, age, temperament, genetic value, debt the farm is carrying, what the auction is paying that month — and the farmer's job is to weigh all of them at once. The farm management software helps. So does experience. So does a glass of bourbon at the kitchen table on a Monday night in early November.

It is easier than farmers admit because, by the time a cow is on the list, the answer is almost always already there. The arithmetic has been pointing one way for weeks or months. The conversation at the kitchen table is not about discovery. It is about acceptance. It is about saying out loud, between two people who have been doing this for thirty years, *Bertha is going. Pearl is going. Daisy is going.* It is about owning it.

Tom does not enjoy this part of the work. He has not enjoyed it in any of the years he has been doing it. He has not, on the other hand, ever stopped doing it.

At nine-thirty Tom walks through the barn. He has not changed his clothes. He has done the morning milking and the pregnancy check on Hazel and made the day's list with Sarah, and now he is doing what he always does on the morning of a cull day, which is the slow walk through the herd that is, in the structure of the day, no different from the walk he does every morning, but is also, on this morning, different.

He stops at Bertha. She is at the bunk. She lifts her head. She looks at him with the steady patient look of a cow who has been on this farm for nine years, who recognizes him, who has, in some Temple-Grandin sense of memory, the experience of him: this person, this voice, this hand. He puts his hand on her flank. She does not move away.

He stands there for a minute. She goes back to eating.

He says quietly, in the low voice that has been the voice he uses with his cows for thirty years: *Vieja. Vieja.* The Spanish was learned from Marcelino over time. Every farmer he has worked with has had a name for the old cows. *Vieja* is one of them. It means *old one*. It is, when said in this voice in this barn at this time of the morning, also a goodbye.

At ten-fifteen the truck arrives.

The truck is owned by a man named Rusty Pelton, who is sixty-four years old, who has been hauling cull cows in upstate New York for forty years, and who has, in that time, transported approximately every fourth cow Tom has ever owned. Rusty drives

a 1998 GMC with a stock trailer behind it. He has a radio in the cab tuned to a country station. He is wearing Carhartt overalls and a Mets cap. He nods at Tom. They have known each other since 1991.

Tom and Marcelino bring Bertha out first. She walks calmly. She has never been a difficult cow. They lead her to the trailer ramp. She walks up the ramp on her own. She steps into the trailer. She looks at Tom over her shoulder in the way cows look at people, which is not in the way humans look at each other but is also not in any other way, exactly. Tom closes the gate behind her.

Pearl comes next. Pearl is more reluctant. She has not been off the farm in five years and the trailer is unfamiliar. Marcelino moves her up with a gentle hand on her flank. She steps in. The gate closes.

Daisy is last. Daisy is, even today, calm.

The trailer holds the three of them. There is room. Rusty signs Tom's paperwork, which has been printed off the herd management software earlier this morning and which Tom hands him on a clipboard. The paperwork tracks the cows to the auction, which will track them to the buyer, which will track them — eventually — to a processing plant in Pennsylvania or western New York. The chain of custody is, by federal regulation, complete.

Rusty hands the clipboard back. He gets in the cab. He starts the truck. He waves out the window. He drives down the lane, past the heifer barn, past the trailer where Marcelino lives, past the maple at the end of the driveway that has lost most of its leaves, and onto the road, and out of sight.

Tom stands at the gate of the lane for a moment after the truck is gone.

Then he turns and walks back to the barn.

The barn that afternoon is quieter than it was the night before. Or it is the same — there are still ninety-one cows in it, and the sound of cud chewing is the same, and the smell is the same, and the light through the eaves is the same — but it feels quieter, to Tom, because he knows three of the cows who were here yesterday are not here today.

Marcelino is in the parlor washing the floor with a hose. Sarah is in the kitchen paying bills. Anna is at Cornell. The dog is asleep under the table.

Tom stands at the back of the barn and looks at the dry pen, where Hazel is, where she has been since the breeding back, eating, growing the calf she will have in August.

The continuity of the farm is a thing he can see, when he stands at this gate, more clearly than at almost any other moment of the year. There were cows here yesterday who are not here today. There are cows here today who will not be here next year. There is a cow in this dry pen now who is carrying a calf who will be born in August, who may stay on this farm for ten years, or may not. There is a heifer in the barn over the hill — Mabel — who is seven months old, who will give her first milk in eighteen months, who may have eight calves of her own over the next decade, who may also not.

A farm is not a fixed thing. It is a slow river of bodies passing through a place, with the place staying constant and the bodies turning over. The work is what holds the place together while the

bodies move. The cows come, the cows go, the calves are born, the heifers freshen, the old cows leave, the milk goes out the door in a tanker, the check comes back on the eighth, and the work continues.

That evening at the kitchen table, Tom is reading the paper and Sarah is on her laptop and the dog is on the rug and the radio is on, low, the way it usually is.

Sarah says: "How was it."

Tom says: "It was OK."

He does not say *Bertha*. He does not need to.

Outside, the November wind is moving across the empty pastures. Inside, the lights are on. In the barn, the herd is settling for the night. In the dry pen, Hazel is lying down, with a calf inside her the size of a small dog, growing on the slow internal clock she has been running on her whole life.

Tomorrow is Wednesday. The milk truck comes Wednesdays.

The work continues.

CHAPTER 11

11

Chapter 11 December The
Year Counted



The first heavy snow of December comes on the ninth, overnight, eight inches between dusk and dawn. Tom wakes up at four to a kind of hushed brightness in the bedroom that snow makes when it has fallen against the window. He puts on the same clothes he puts on every morning. He goes downstairs. He starts the coffee. He goes outside.

The path to the barn has to be plowed before he can do anything else.

He gets the skid steer out. He runs the bucket down the path. The light from the milkhouse is yellow against the white. Marcelino is already in the parlor — Marcelino is always already

in the parlor — and the cows are already standing at the holding pen gate, blinking in the bright light at four-fifteen in the morning, in the first deep cold of the year, ready to be milked.

December has arrived.

Bill Moriarty is the farm's accountant. He has been the farm's accountant for twenty-two years. He inherited the work from the man who had been the farm's accountant before him, a man named Frank Moriarty, his father, who had worked with Tom's father since 1981. Bill came to the farm for his first year-end review in 2003. He has come every year since. It is the second Tuesday of December.

He arrives at one in the afternoon. He drives a Subaru. He is fifty-eight. He brings a thick manila folder and a laptop and his own bag of cookies because he is on a diet and Sarah's cookies are a problem. He sits at the kitchen table with Tom and Sarah and the laptop and the bourbon Tom brings out for the occasion. They will be at the table for three hours.

This visit is the closest thing the farm has to an annual physical. Bill tells them what the year looked like. What was earned. What was spent. What the depreciation schedule is. What the federal forms will say. Where the year stands relative to the year before, and the year before that. He is the only person besides Sarah who sees the farm's complete financial picture, and unlike Sarah he sees it from outside the kitchen.

The year, in summary form: gross milk income \$537,400, slightly down from 2024 because of a soft October. Operating expenses \$341,200, slightly up from 2024 because of feed costs and one major equipment repair (the bulk tank refrigeration). Net

before taxes and family living: \$196,200. The drought reserve account, which was drawn down in 2022, is back to roughly half full. Anna's tuition for the year was paid. The family's medical insurance was paid. The mortgage was paid. The new water heater was bought.

"You made it," Bill says.

"That's what it looks like."

"You did better than you think."

"What does better than I think look like?"

Bill turns the laptop a little. He shows them the chart he has made. The chart shows their net farm income for the past ten years. There are years above the trend line and years below. 2025 is just above. 2022, when the drought hit, is far below — a deep notch. 2018, the year they bought the eighty-acre piece, is a notch in the other direction. The trend line is more or less flat.

"Flat is good in this business," Bill says. "Flat is good."

The federal government does not run American agriculture, but it is in the room.

The Farm Service Agency is the part of the U.S. Department of Agriculture that administers most of the federal programs that touch a farm like Tom's. There is the Dairy Margin Coverage program, which is a federal insurance program that pays out a small amount per hundredweight when the difference between the national milk price and the national feed cost falls below a threshold. There are the Environmental Quality Incentives Program and the Conservation Stewardship Program, which fund things like fencing for rotational grazing, cover crops, manure management, soil health practices. There are crop insurance

programs. There are disaster relief programs. There are loan programs for farmers who cannot get credit from banks. There are dozens of forms.

Tom is enrolled in the Dairy Margin Coverage program. He gets a small payment from it most years. He has used the conservation programs for fencing and a stream-buffer project. He files the forms. He tries to remember the deadlines. Sarah, who has been filing federal paperwork for thirty years, makes sure they get filed.

This is part of how a small dairy stays a small dairy. The federal programs are not generous. They are not transformative. They are not, in any sense, a solution to the structural problems of small dairy in America. But they are something. They are a thumb on the scale, in years when the scale tips far enough, that keeps a farm like Tom's from going under.

Without them, the farm would not exist in this form. Bill knows this. Tom knows this. The forms get filed.

There are things the spreadsheet does not capture.

It does not capture that Bertha is not in the herd anymore. The spreadsheet shows a smaller herd average for November and an "asset disposal" line for \$1,840, which is what the auction paid for her at New Berlin. It does not show the place she had walked to first on every turnout day for seven Aprils, which is now a place that no cow walks to first because no cow remembers it the way she did.

It does not capture that Marcelino is still here. The spreadsheet shows a labor expense of \$58,700, which is up from last year because Tom raised his pay. It does not show that on the morning

of November 14, when Tom was walking back from the truck, Marcelino was the one who finished the floor of the parlor without being asked.

It does not capture that the equipment is older. The spreadsheet has a depreciation schedule. The depreciation schedule is a stylized thing the IRS uses to allow farmers to deduct the cost of capital. It is not, in any literal sense, the actual aging of the actual machinery. The skid steer that Tom is using to plow the path this morning is twelve years old. It is going to need a new hydraulic pump in 2026 or 2027. The bulk tank refrigeration that was repaired in 2023 is not the same unit it was before the repair, and it makes a sound at night, when Tom is in the milkhouse late, that he does not entirely trust.

The spreadsheet captures what the spreadsheet captures. Some of what matters about a farm in any given year does not show up there.

After Bill leaves, around four, Tom puts on his coat and walks the barn one more time before evening milking. This is a habit. He has been doing this on the day of the accountant's visit since he was thirty.

The herd is ninety-one. There are twelve heifers who will join the milking string next year. There are eight more who will join the year after. Mabel is in the older heifer group, in the back of the heifer barn, standing at the bunk eating with the other nine-month-olds. She is half-grown. She is a brown-and-white animal with the long legs of a calf still and the chest beginning to fill in toward the cow she will become. Tom stops at her pen. She lifts her head. She does not, exactly, recognize him; she has not had enough

interactions with him for that. But she does recognize the shape of a person at the gate, which means feeding might be about to happen, and she takes a step toward him, in case.

He moves on. Hazel is at the bunk in the milking barn, eating, her belly still mostly the belly of a non-pregnant cow but slightly fuller than it was in September. Inside her, the calf is the size of a kitten. Tom pats her shoulder as he goes by. She does not turn.

The herd is the herd. The cycle is turning.

Anna comes home from Cornell on the eighteenth and stays through the second of January. She is a senior now. She is talking, vaguely, about graduate school in soil science, and she is talking, less vaguely, about a young man in her department whom she has been seeing for three months, and Tom is listening to all of this with the careful neutral face of a father trying not to be a problem. Sarah is more direct. Sarah likes the young man, whom they have not met. Anna says they will see each other at Christmas if it works out, which has not yet been determined.

The Christmas tree goes up on the twenty-first. Sarah has been putting up the same Christmas tree for twenty-eight years; it is not a real tree, it is an artificial one that came from a Sears in 1996 and which Sarah has periodically threatened to throw out and never has. The ornaments are mostly homemade. Anna's first-grade clay handprint is among them. So is a small wooden cow that Tom's grandfather carved.

Marcelino is in his trailer. He comes to the house for Christmas Eve dinner, as he has every year for five years, sitting in the seat at the dining room table that has been his on Christmas Eve since 2020. He brings a bottle of Guatemalan rum that he has saved for

this. They eat ham and Sarah's pierogis (she is half Polish from her mother's side) and a green bean casserole that nobody especially likes but is, somehow, a tradition. They give him a card with three hundred dollars in it, which is the same gift, adjusted for inflation, that Tom gave the man who lived in the trailer before Marcelino. He says *Gracias*. He says *Feliz Navidad*. He goes back to his trailer at nine, with leftovers wrapped in tinfoil.

At eleven that night his phone rings. It is Marisol. It is nine in Guatemala. She is opening the present he sent — a doll with curly hair that he ordered from a website in October and had shipped to the house in San Pedro Necta — and she is screaming with delight.

¡Papi! ¡Papi! ¡Mira! ¡Mira!

The phone is on speaker. The trailer is small. The doll, on the screen, is held up by a seven-year-old in pajamas. The doll, like Marisol, has dark hair. Marcelino watches his daughter for as long as she will let him, which is several minutes, and then she is off — to show her grandmother, to show a cousin, to do something important with her present that does not involve the phone — and the phone screen is lying on a table in a kitchen four thousand miles away, and Marcelino is sitting on the edge of his bed in upstate New York at eleven o'clock on Christmas Eve with his eyes closed.

In a few minutes Lucía comes back to the phone. They talk for a while.

He sleeps.

The year ends. The cows are milked. The snow falls. New Year's Eve passes without much ceremony at the farm, because four o'clock comes early on January first. The Christmas tree comes down on the third. Anna goes back to Cornell on the second.

By the second week of January the farm has begun the deep winter rhythm that it will be in for the next six weeks. The cows are inside. The calves are in the hutches. The trailer behind the barn glows yellow in the dark at five in the morning. The milkhouse glows yellow at four. The skid steer scrapes the alleys. The vacuum pump runs.

Mabel grows. Hazel grows the calf inside her, who is now four months along and the size of a small cat. The herd settles into the long stretch between the year that just ended and the year that has just started, and the work goes on, without ceremony, the way the work has gone on through every December and into every January in this barn and on this farm for as long as anyone here remembers.

In a few weeks Tom will be in the kitchen at four in the morning, putting on the second of his two pairs of socks, and Marcelino will be in the milkhouse, and the dry cows will be in the pen at the back of the barn. One of them will be Hazel. She will be six months along. Her calf will be the size of a small terrier. In two more months she will calve.

This is the wheel. This is what it does.

It does not stop.

CHAPTER 12

12

Chapter 12 Could You Do
This



You have, almost certainly, thought about it.

Most readers of this book have, at some point in the last few years, imagined some version of the following. A farmhouse somewhere — Vermont, maybe, or western Massachusetts, or upstate New York, or the Pacific Northwest. Ten acres, more or less. A barn. A garden. Chickens. Maybe a couple of goats. Maybe a milk cow. A wood stove. A view. A partner who is in on the project. Children who grow up free of screens, with their hands in the dirt. A small income from the farm, or a remote job that lets you live this way. A life that is more honest, more grounded, more real than the one you grew up in.

This vision is real to you. You have seen it, in one form or another, in films, in Instagram posts, in cookbooks, in the lives of writers and homesteaders whose work you follow. You have, possibly, said something like this out loud to a friend.

The vision is not, exactly, wrong. People do live this way. Some of them are happy. Some of them are even solvent.

It is also not what you think it is.

The honest answer takes more than one sentence to give, but it begins with this: most people who try this, fail. The reasons they fail are specific and worth understanding. And the people who succeed at it are, much more often than not, people whose lives have prepared them for it in ways that are not obvious from the outside.

Let us do some arithmetic.

Ten acres of land in a part of the country where you would actually want to live — within driving distance of jobs and hospitals and grocery stores, in the climate zones that support the kind of farming you imagine — costs, in 2025, somewhere between fifty thousand dollars and three hundred thousand dollars for the bare land. The wide range is because land prices vary enormously: rural northern Maine is cheap, the Hudson Valley is expensive, Vermont is in between, the Berkshires are higher than Vermont, and a few miles can change the price by a hundred percent. For ten useful acres in a place you would want to live — call it the Berkshires or the Finger Lakes or central Vermont — figure something like a hundred and twenty thousand dollars for raw land.

But you cannot live on raw land. You need a house. Depending on the property, you might find a small farmhouse already on the land for an additional hundred to two hundred and fifty thousand dollars, or you might need to build, which will cost three hundred thousand or more. You will also need a barn or some kind of outbuilding, which is another fifty to a hundred thousand. You will need fencing, which is more expensive than people think — figure two thousand to three thousand dollars per acre to fence properly. You will need water for animals, which means wells or hydrants or both. You will need machinery, even if you go small — a tractor, a truck, a chainsaw, a generator, smaller hand tools — and you will spend at least thirty thousand on equipment to start.

The all-in cost of getting onto a ten-acre farm in a desirable region of the United States, with a house and the basic infrastructure to operate it, in 2025, is somewhere between four hundred thousand dollars and seven hundred thousand dollars.

This is not money you can borrow easily. Banks do not lend against farms the way they lend against suburban houses. The USDA's farm loan program exists and is helpful for some buyers, but it has restrictions and waiting lists. Most people who get onto a small farm fund the purchase with one of three things: family money, money saved from a previous career, or the sale of an urban or suburban home. This last one is how most of the people who write the books and run the Instagram accounts about small farming actually got started — they sold a house in Brooklyn or Boston or San Francisco and used the equity to buy land. This is not nothing. It is a real thing. It is also not available to a person who is starting from zero.

Then there is the work.

A working ten-acre operation, even on a small scale, requires somewhere between forty and eighty hours a week of physical labor from the people running it. The high end is during planting season and harvest and haying. The low end is in the depths of winter when there is nothing growing. The animals — and you will have animals, because vegetable-only farming on ten acres is even harder economically — will need to be fed every day, watered every day, managed for health every day, and dealt with when they are sick or dying or giving birth, which they will be, often. There are no weekends. There are no vacations. There is, in fact, in the early years, almost nothing else.

The work is also, by the standards most readers of this book are used to, physically punishing. It involves lifting things that are heavy. It involves working in cold and rain and heat. It involves bending, kneeling, dragging, hauling. It involves your hands being in dirt and water and animal mess and your own sweat for several hours every day. It involves the kind of cumulative repetitive strain that, over years, will show up in your back and your knees and your wrists. None of this is, by itself, a reason not to do it. People do this work. People love this work. It is, however, a kind of work most readers of this book have never been asked to do for a sustained period, and it is a kind of work that requires a particular type of competence that takes years to develop.

You will also be, much of the time, alone. The romance of small farming includes other people — a partner, a community, a network of friends — but the actual hours of the work are mostly hours of one person doing physical things in a field or a barn, by themselves, for stretches of three or four or six hours at a time. A reader who has spent most of their life in school, in cities, around

peers, in conversation, may not know in advance how their mind responds to this kind of solitude. Some people are nourished by it. Some people are slowly hollowed out by it. You cannot, easily, predict which one you will be.

The people who succeed at this kind of life, in the United States in the twenty-first century, are not a random sample of those who try.

They are, overwhelmingly, one of three kinds of people.

The first kind are people who grew up on a farm. They watched their parents do it. They know, in their bodies, what the work is. They know what equipment is, what animals are, what weather is, what a bad year feels like. They are not surprised by the long hours or the physical toll because they have seen it their whole lives. Most successful small farmers in America today come from farm families. This is not a story about heredity. It is a story about apprenticeship — about the long years of unconscious learning that happen when you are a child watching adults do work you will eventually do yourself.

The second kind are people who married into a farm family. The dynamics are similar. They had, before they did this, a partner who knew how. They had a family that taught them. They had a way in.

The third kind are people who came to it as adults, but who came with two specific things: enough capital to absorb several years of losses, and enough humility to apprentice themselves to people who already knew how. The most successful adult-entry small farmers I have read about all spent at least two years working on someone else's farm before starting their own. They

did not buy land first and figure it out. They learned the craft first, slowly, on someone else's nickel, and then bought land, and then made the mistakes you can only avoid by having already made them.

The people who fail — and most of the people who try, fail — are usually some version of the same story. They had the dream. They had some money, but not enough. They had no farming background. They did not apprentice. They bought the land and tried to learn on the job. The first year was hard but exciting. The second year was hard and not exciting. The third year was hard and lonely and broke. Sometime in the third or fourth year, almost always, something happens — a marriage strain, a financial collapse, a serious injury, the death of an animal that the owner cannot bring themselves to face — and they sell. The land goes back on the market. Someone else buys it. The cycle continues.

A specific word for the reader of this book.

You are, statistically, smart. Your education has been good. Your parents have arranged your life carefully and, on the whole, lovingly. Much of what that arrangement has accomplished is to protect you from physical labor, from discomfort, from failure with your hands. These are not bad protections to receive. They are also not the protections that prepare a person for small farming.

Small farming requires a competence with one's body, and with the slow ungainly mistakes physical work demands, that has to be built — by years of doing the work, by tolerating the early failures, by accepting that there is a long stretch in which you are bad at the thing you are trying to learn. Most readers of this book have not been asked to spend any sustained period being bad at something

physical. The schools you have gone to do not, mostly, teach this. Your parents, however much they love you, do not, mostly, model it.

This is not a reason not to try, if you want to try. It is a reason to be honest about where you are starting from. You are not starting from zero. You are starting, in some respects, from below zero, because you have to unlearn an unconscious expectation that work yields to effort and that effort yields to talent, neither of which is reliably true in farming. The first year, for almost everyone in your position, will be humiliating. The second year, less so. By the fifth year, you will know whether this life is for you.

There is no faster path to that knowledge than five years.

This is not new. Americans have been doing this for a long time.

There were waves of back-to-the-land settlement throughout the twentieth century, and they followed roughly the same arc. The largest and most studied was the wave of the late 1960s and early 1970s, when several hundred thousand mostly young, mostly white, mostly college-educated Americans left cities for rural land — Vermont, Tennessee, New Mexico, northern California — to start farms or communes or homesteads. Many of them carried specific ideological commitments: anti-consumerism, environmentalism, skepticism of urban industrial society, an interest in self-sufficiency.

Most of them, within five to ten years, were gone.

The reasons varied. Some discovered they could not handle the work. Some discovered they could not handle their partners or their communes. Some discovered the romance of poverty wears

thin around the second winter without heat. Some discovered, having left the city to escape conformity, that rural communities had their own forms of conformity, and that being a long-haired stranger from Cambridge did not always read locally as a free spirit. Some discovered that their bodies were not as durable as they had thought. Some had children, who required schools, healthcare, social lives, and could not all be raised on lentils and herbal tea.

The ones who stayed, again, fell into the categories described above. They had local family. They had outside money. They had real apprenticeship. They were, in a few cases, just exceptionally stubborn and exceptionally lucky.

What this period left behind, beyond a generation of people with a story about that one year in Tennessee, was a set of practical institutions — small farm programs, organic certifications, sustainable agriculture research networks — that exist today and which are useful, in real ways, to anyone who wants to try this. The wave failed at its individual ambitions and succeeded at building infrastructure. The reader who eventually goes onto a small farm will be standing on top of that infrastructure, whether or not they know it.

A particular kind of person is now telling the story of small farming, on Instagram and on television and in books, in a way that has begun to feel like the dominant version of the story.

These people are not, generally, lying. They are showing parts of their lives. The parts they are showing are real. What is missing, often, is context. The young couple in the renovated farmhouse with the immaculate garden and the photogenic chickens has, in many cases, an inheritance, a successful spouse with a remote tech

job, a book deal, a brand partnership, a substantial Instagram income, a partial retirement from a previous career, or some combination of the above. The labor in the photos is real. The land is real. The way they came to be there is, often, not what the photos suggest.

This is not a moral failing of theirs. They are not obligated to put their tax returns on Instagram. But it is useful, when looking at this kind of content, to remember that what makes the life sustainable for them is, often, not what is being shown. The aesthetic and the economics are separable. The aesthetic can be reproduced almost anywhere by a person with means; the economics of small farming are what they are, regardless of how a sunset looks on the chicken coop.

If you are going to use the work of any of these creators as inspiration, use it. There is real value in seeing a life that is different from your parents'. But understand, before you make decisions, what is and is not visible in what you are seeing.

There is another version of this worth knowing about, although it does not exactly answer the question this chapter has been asking.

The example comes from the memoir of Hank Paulson, who served as Treasury Secretary of the United States during the 2008 financial crisis. Before he was Treasury Secretary, before he was the chief executive of Goldman Sachs, before any of his career had happened, he was a young man recently married to a young woman named Wendy. They had met at college. He had grown up on a farm in Barrington Hills, Illinois, outside Chicago, where he had milked cows as a boy before school — a thing he writes about in the memoir at greater length than he writes about most things,

the way some men in his industry write about prep school. The two of them, when they were starting their lives together, did not move to an apartment in New York or Chicago to be near his work. They bought a piece of his parents' farm, and they built, themselves, by hand, the first home they ever shared. For many years it was a loft. Wendy did the wiring. Hank, by his own admission, was clumsy with his hands, and let her.

They did not, after that, leave. The career happened around the house. He commuted, often long distances, sometimes by plane, sometimes only on weekends. The house was where they came back to. The land was where their children grew up. The job, however demanding, however prestigious, was the thing that was elsewhere. The home was the farm.

This is a lifestyle choice that almost no one in the audience of this book has been told they have. The default story most readers have absorbed is that a serious career means living near it — in the city, in the suburb of the city, in the apartment with the doorman. The notion that you could put your stake in a piece of rural ground when you are twenty-five, build a loft on it with your own hands and your spouse's, finish the wiring on weekends because someone has to, and keep that as the place where you live for the next fifty years while a career happens somewhere else — this is not the story that usually gets told. But Hank and Wendy Paulson did it. So have other people. The lives they built are not, on the whole, lives they regret.

This chapter cannot tell you whether such a life is the right one for you. That is a thing you will figure out, slowly, as you go. But it is a thing worth knowing — that the choice between *farmer* and *not*

farmer is not the only choice. There is also the choice of *farmer-rooted*, in which the work of your life is not farming but the place of your life is the farm.

If, having read all of this, you still want to try — and some of you will, and a few of you should, and the world will be better for the few of you who do it well — then a sensible smaller-scale entry point is goats.

Goats are not cows. Goats can be kept on smaller acreage. Goats can be handled by a single person without specialized equipment. Goats are forgiving of mistakes that a cow would not be forgiving of. Goats can be milked by hand. The capital requirements for a small goat dairy are a fraction of those for a small cow dairy. Goats produce a value-added product (cheese, soap) that can be sold directly to consumers at farmers markets, which avoids the commodity-pricing problem that destroys most small cow operations.

Goats are also, it should be said, mischievous. They escape. They eat things they should not eat. They have personalities. They get into fights. They will, on the day you have an important meeting in town, find the one weak spot in the fence you did not know was there. None of this is a reason not to keep goats. It is, however, the reality of keeping goats, and you should know it before you commit.

A small goat operation — twenty animals, two acres of pasture, an outbuilding, basic milking equipment, a small certified kitchen for making cheese — can be entered on a budget of less than a hundred thousand dollars. It will not, by itself, replace a salary. It can, in combination with a part-time job or a remote income or a working partner, supplement a household. It can, over several

years, become a real business. The path to viability is harder to find than it sounds, but easier than the cow-dairy path, and it is something a person without a farming background can plausibly learn.

There are people who have done this. There are books written by them. The books are useful.

If you want to do this, or if you want to even think about doing it, here are the resources that actually help.

The first is your state's Cooperative Extension Service. Cooperative Extension is a nationwide network of land-grant university programs that provide free or low-cost agricultural advice, classes, soil testing, and consultation to anyone who asks. Cornell Cooperative Extension, in New York, runs a Small Farms Program that has guided thousands of beginning farmers through the practical questions of starting a small operation. Every state has an equivalent. You can call them. You can take their classes. You can attend their field days. They are, in most cases, staffed by people who are good at this and who are paid by the state to help you.

The second is the federal Sustainable Agriculture Research and Education program, known as SARE. SARE funds research on sustainable farming practices and publishes a substantial library of free guides on everything from rotational grazing to small-scale dairying to financial planning for small farms. The guides are not exciting. They are correct.

The third is your state's small-farm advocacy organizations. New York has the Northeast Organic Farming Association. Vermont has the same plus several others. California has CCOF.

Most regions have one. They run conferences, mentorship programs, apprenticeship networks. They are run by people who are trying to keep small farms alive and who can, often, connect you to a working farm willing to take an apprentice for a season.

The fourth is the people themselves. Most working small farmers will, if approached politely and with seriousness, talk to a beginning farmer for an hour. Some will let you work on their farm. Some will become mentors. The American small-farming community is small enough that most of the serious people in any region know each other. If you are willing to apprentice, with humility, for one or two or three years before starting your own operation, doors will open for you that will not open for someone who buys land first.

This chapter is not trying to talk you out of it.

There are reasons to live this kind of life that have nothing to do with money or with whether you will succeed at the business of farming. There are real things this kind of work gives a person — a relationship with land, a relationship with animals, a relationship with weather and seasons, a kind of competence that almost nothing else in modern American life delivers — that you cannot get any other way. The fact that the project is hard and that most people who attempt it fail is not, by itself, an argument against trying. Many things worth doing are like this.

What this chapter is trying to say is something a little more specific. It is trying to say: if you are going to try this, try it with clear eyes. Know what it costs. Know who succeeds. Know why most fail. Know that the path with the best chance of working is the slow one — the apprentice, the second job, the years of learning — and not the fast one. Know that the version of this life

you have seen on Instagram is not the version most people doing it actually have. Know that the people for whom small farming is not a choice — the people whose families have been doing it for generations, in poverty, around the world — would mostly trade places with you in an instant.

And know that the system that produces the food on most American tables, the system this whole book has tried to describe, runs on a small number of farms and a smaller number of farmers and a kind of work that no one is currently lining up to do. There is a real shortage of young people willing to milk cows and fix fences and feed calves at four in the morning. The country needs them. If you are going to be one of them, the country is grateful.

The country will not, mostly, pay you what you are worth.

That is the truth, told as honestly as one chapter can tell it.

CHAPTER 13

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Chapter 13 Coda



Hazel calved on the eighth of August. It was a heifer, brown-and-white, born at four-fifteen in the morning to no particular weather. Tom told me about it on a phone call. The calf had been an easy birth. The calf was healthy. Hazel was fine. Sarah, in the background of the call, said something I couldn't quite hear, and Tom laughed.

I am writing this in November of the same year, in a city, in a building, three hundred miles from the farm. Outside my window the trees are bare and a person I do not know is walking a dog past a parked car. There is no animal in this room. There is no field. The cheese in my refrigerator came from somewhere I have not been.

This is the ordinary geography of most of our lives. We are, almost all of us, on the consuming end of a system whose producing end we do not see. The visible part of our food economy is a refrigerated case, a register, a paper bag. The invisible part is everything that came before it. This book has been a small attempt to push back against the invisibility — to put one farm, one farmer, one worker, one cow, on the page in enough detail that the reader carries something they did not have before.

What did the year teach me?

A few things, none of them tidy.

That the cow is a creature. I had not, before I started this book, spent meaningful time around dairy cattle, and I had thought of them — without examining the thought — as something halfway between a mammal and a thing. They are not. They are large warm specific animals, with friends and enemies and routines and memories, who watch you with calm attention and chew their cud and live the lives we have shaped them to live. To stand in a barn at four in the morning and feel the breath of ninety-four bodies in the dark is to feel a kind of presence that is not human and is not nothing. Once you have felt it, you cannot un-feel it. It changes how you read the carton on the shelf.

That the work is harder than I thought. I am not a person of soft hands, but I had not, until this year, understood the cumulative cost of a kind of labor that does not stop. Tom is fifty-six and his body is older than that. Marcelino is twenty-nine and his body is, in some ways, also older than that. The toll is not abstract. It is shoulders. It is knees. It is the specific knot in the lower back that returns to a person every morning of their life. The food on most American tables is built, in part, out of this toll. The

toll is mostly invisible, and mostly uncompensated, and is borne by people who are mostly not in the rooms where their lives are being decided.

That the system is not anyone's fault, and is also a fault. Tom is a farmer doing his best inside a structure he did not build. Marcelino is a worker doing his best inside a legal regime he did not write. The retailer in Boston is selling a carton at the price the market will bear. The shopper is paying that price and going home. No single person is to blame for the shape of the whole. The shape of the whole is, nonetheless, what it is. Whether anyone has the responsibility to change it, and what that responsibility might require, is not a question this book has tried to answer. It is a question worth asking, slowly, over years.

That the people who do this work love it, mostly. Tom loves his cows. Marcelino loves his daughter, and is doing this work because of her. Sarah loves the place, and the rhythm of the year, and the book she keeps in pencil that says where every cow is and what every cow has done. The work is hard and the pay is bad and the schedule is unforgiving. None of that is the whole story. The whole story includes what the work also is — a way of being in the world that some people have chosen and some people have been chosen by, and which, on the days when it is going well, is unlike anything else.

And that the wheel keeps turning. The book began in February with a cow about to calve. The book ended in December with that cow carrying the next calf inside her. By the time this book is in your hands, that calf — Hazel's August calf — will be six months old, and she will be in the heifer barn, growing into the cow she will be. By the time you are old enough to remember reading this,

she will be in the milking string. By the time your children are old enough to read it, she will be gone. Mabel will be older. Hazel will be older. The farm will still, with luck, be there. The work will still be done. The milk will still go out the door.

This is what a year teaches a person. That the world we live in is held together by quiet, daily, unromantic acts performed by specific people in specific places, and that those people and those places are visible if you look for them, and worth looking at.

The book is over. The farm is not.

CHAPTER 14

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Chapter 14 Sources And Further Reading



The notes below are not academic citations. They are recommendations, organized by chapter, for any reader who wants to keep going. I have stayed mostly with sources that a serious fifteen-year-old can actually read and that are, in most cases, available at a public library or online.

A few books appear under more than one chapter, because the best books on this subject tend to cover ground in several of them at once. The most generally useful three, if you only ever read three, are Verlyn Klinkenborg's *More Scenes from the Rural Life*, Anne Mendelson's *Milk: The Surprising Story of Milk Through the Ages*, and Margaret Gray's *Labor and the Locavore*.

Chapter 1 — February: Inside the Barn

Verlyn Klinkenborg, *More Scenes from the Rural Life* (NYRB, 2013) — short essays on farm life in upstate New York, written with the kind of patient attention this book has tried to imitate.

Wendell Berry, *Bringing It to the Table: On Farming and Food* (Counterpoint, 2009) — a small selection of Berry's farming essays, more accessible than his longer works.

Catherine Friend, *Hit by a Farm: How I Learned to Stop Worrying and Love the Barn* (Da Capo, 2006) — a funny memoir about adult-entry small farming. Useful as a counterweight to the romanticism in this genre.

Anne Mendelson, *Milk: The Surprising Story of Milk Through the Ages* (Knopf, 2008) — the best single book on the cultural and biological history of dairy.

Chapter 2 — The Cow Herself

Temple Grandin and Catherine Johnson, *Animals in Translation* (Scribner, 2005) — Grandin's most accessible book, and the one that explains how cows actually perceive the world.

Temple Grandin, *Thinking in Pictures* (Doubleday, 1995) — Grandin's autobiography. Worth reading even if you do not care about cows, but especially if you do.

Laurie Winn Carlson, *Cattle: An Informal Social History* (Ivan R. Dee, 2001) — a readable history of how humans and cattle have lived together for ten thousand years.

Anne Mendelson, *Milk* (above).

Chapter 3 — March: A Calf Is Born

Sonja A. Swanson and Kevin J. Anderson, *Cow-Calf Separation in Dairy Production* — a 2021 review article in the *Journal of Dairy Science*. Free online. Summarizes the research on what separation costs cow and calf.

Marc Bekoff, *The Emotional Lives of Animals* (New World Library, 2007) — a careful book on what we know and do not know about how animals experience their lives.

Jonathan Safran Foer, *Eating Animals* (Little, Brown, 2009) — the most-read recent book on the moral side of animal agriculture. Disagree with it where you want to. Read it.

Chapter 4 — April: Out to Pasture

Allan Savory, *Holistic Management* (Island Press, 3rd ed. 2016) — the canonical book on managed grazing. Worth reading, with the caveat noted in the chapter.

American Grassfed Association, americangrassfed.org — the certification standards in full, plus directories of certified producers.

USDA Organic Standards, ams.usda.gov/rules-regulations/organic — the federal organic rules, which are public and surprisingly readable.

Sarah Flack, *The Art and Science of Grazing* (Chelsea Green, 2016) — the best practical guide to rotational grazing for small farmers.

Judith D. Schwartz, *Cows Save the Planet* (Chelsea Green, 2013) — accessible introduction to soil science and grazing's role in carbon cycling. Reads its claims with the appropriate critical attention.

Chapter 5 — The Morning

John Smith and Mike Brouk, *Milking Systems and Parlors* — Penn State Extension publication, free online. The clearest short explanation of how milking machines actually work.

Robert Wolke, *What Einstein Told His Cook* (Norton, 2002), chapter on dairy — useful background on the chemistry of milk and how it gets from cow to carton.

John Bunting, *The Milkweed* (newsletter, archives at themilkweed.com) — Bunting was a dairy farmer who wrote pointedly about the industry for decades. Useful for the working farmer's perspective.

Chapter 6 — June: The Milk Check

USDA Agricultural Marketing Service, *Federal Milk Marketing Order Statistics* (ams.usda.gov) — the data behind the Federal Milk Marketing Order system, updated monthly.

Marin Bozic et al., *Risk Management for Dairy Farms* — University of Minnesota extension publications. Free online. The clearest explanations of how dairy pricing actually works.

Brenda Cooper, *Milked: How an American Crisis Brought Together Midwestern Dairy Farmers and Mexican Workers* (The New Press, 2024) — a recent book that combines the economics of small dairy with the labor question. Useful for both this chapter and chapter 7.

USDA NASS, *Milk Production* annual reports — for the long-run data on the consolidation of American dairy.

Chapter 7 — August: Haying and the People Who Do It

Margaret Gray, *Labor and the Locavore: The Making of a Comprehensive Food Ethic* (University of California Press, 2013) — the indispensable book on this subject. If you read one book about farm labor, read this one.

Brenda Cooper, *Milked* (above).

Seth Holmes, *Fresh Fruit, Broken Bodies: Migrant Farmworkers in the United States* (UC Press, 2013) — about fruit and vegetable workers, but the analysis applies broadly.

Migrant Justice / Justicia Migrante (migrantjustice.net) — a Vermont-based farmworker advocacy organization. Their reports are useful primary sources.

Jonathan Blitzer, *Everyone Who Is Gone Is Here* (Penguin Press, 2024) — for context on Central American migration to the United States, including the conditions that have driven many dairy workers north.

Chapter 8 — Goats, Sheep, and the Cheese Across the Ocean

Comté Cheese Association, comte.com — the official organization. Their public materials are detailed and informative.

Pierre Boisard, *Camembert: A National Myth* (UC Press, 2003) — about a different French cheese, but the best English-language book on how AOC systems actually work.

Jenna Woginrich, *Made from Scratch* (Storey, 2009) — a memoir of small-scale homesteading that includes goats. Honest about the difficulty.

Gianaclis Caldwell, *Mastering Artisan Cheesemaking* (Chelsea Green, 2012) — if you actually want to make cheese, start here.

The Cellars at Jasper Hill, jasperhillfarm.com — a Vermont operation that has built something Comté-like in the United States. Their model is worth understanding.

Chapter 9 — October: Breeding Back

Select Sires, selectsires.com — the catalog from one of the major dairy genetics companies. Worth seeing for yourself how bulls are marketed.

Heersche, Hopkins, and Spire, *Reproductive Anatomy and Physiology of the Cow* — Kansas State Extension publication, free online. Clear and accurate.

Helen Browning and Tim Finney, *What's Wrong with Modern Dairy Cattle Breeding? An Organic Perspective* — short paper from the Soil Association (UK), free online. Critical perspective on industrial genetics.

Chapter 10 — November: The Decision

Bernard E. Rollin, *Farm Animal Welfare: Social, Bioethical, and Research Issues* (Iowa State Press, 2003) — the philosophical literature on farm animals, written by a philosopher who has spent his career thinking about this carefully.

USDA, *Dairy Cattle Disposition Reports* — the public data on how cull cows enter the beef supply chain.

Verlyn Klinkenborg, *Several Short Sentences About Writing* (Knopf, 2012) — not about cows. Listed here because the chapter on culling owes its restraint to Klinkenborg's instructions on how to write quietly about hard things.

Chapter 11 — December: The Year, Counted

USDA Farm Service Agency, fsa.usda.gov — the agency itself, including descriptions of all the programs mentioned in the chapter.

Cornell PRO-DAIRY, prodairy.cals.cornell.edu — the financial benchmarking reports they publish are the closest thing to public data on what actual New York dairy farms earn.

National Sustainable Agriculture Coalition, sustainableagriculture.net — for advocacy and policy context on the federal programs that touch small farms.

Chapter 12 — Could You Do This?

Cornell Small Farms Program, smallfarms.cornell.edu — start here if you live in or near New York. Every state has an equivalent through its Cooperative Extension service.

SARE (Sustainable Agriculture Research and Education), sare.org — the federal program. Their *Building a Sustainable Business* is a free workbook that walks a beginning farmer through real planning.

Northeast Organic Farming Association (nofa.org) and equivalent organizations in other regions — for conferences, mentorship, and apprenticeship networks.

Eleanor Agnew, *Back from the Land: How Young Americans Went to Nature in the 1970s, and Why They Came Back* (Ivan R. Dee, 2004) — a book by a participant in the back-to-the-land movement, looking back honestly. The clearest account of why most attempts ended.

Henry M. Paulson Jr., *On the Brink: Inside the Race to Stop the Collapse of the Global Financial System* (Business Plus, 2010). Most of this memoir is about the 2008 financial crisis. But the early chapters contain something this book has tried to gesture at — the story of a young couple who, in their twenties, chose to start their married life in a loft they built themselves on a piece of his parents' farm in Illinois, and never left, even as his career took him to Goldman Sachs and the Treasury. The wiring was Wendy Paulson's. The clumsiness was Hank's. Read the first few chapters.

Helen and Scott Nearing, *Living the Good Life* (1954) and *Continuing the Good Life* (1979) — the original American homesteading books. Worth reading both for inspiration and as a check on what their lifestyle actually required (a partial pension, no children, very particular philosophical commitments).

Wendell Berry, *The Unsettling of America* (Sierra Club Books, 1977) — the foundational American argument for small-scale agriculture as a moral and ecological practice. Forty-eight years old and still the book to read.

Michael Pollan, *The Omnivore's Dilemma* (Penguin, 2006), and the young readers' edition (Dial, 2009) — the most-read recent book on where American food comes from. The young readers' edition is the model many of the choices in this book have, in their own way, tried to follow.

A Note on What Is Not Here

I have not included most of the academic literature on dairy science, animal welfare, or food systems policy, although these literatures are large and worth knowing about. A reader who wants to go further into any of these can start with the bibliographies of the books listed above, most of which lead into the academic conversation.

I have also not listed every farm memoir, every documentary, or every newspaper article on the subject. These exist in abundance. The recommendations above are the ones I would, if asked by a younger version of myself who wanted to understand what an American dairy farm is, hand over without hesitation.

The list is not closed. Other readers will bring other books to it. That is how it should be.